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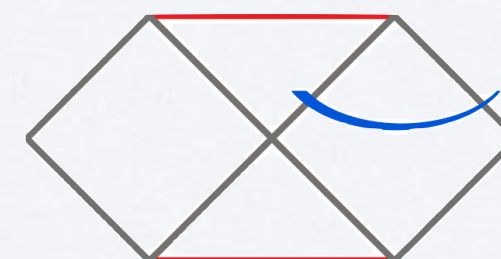


TIME REVERSAL ‘SYMMETRIES’ **IN THE** **HYPERBOLOIDAL FRAMEWORK**

Rodrigo Panosso Macedo

Marica Minnuci, Laura Sberna, Christiana Pantelidou

[Arxiv gr-qg: 2604.13182]



<https://hyperboloid.al>

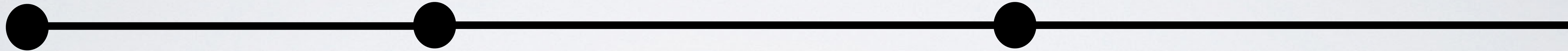


OUTLINE

QNM products

$t-\phi$ 'Symmetry'

Future/past hyperboloidal

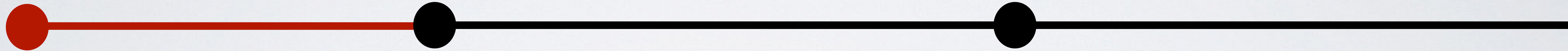


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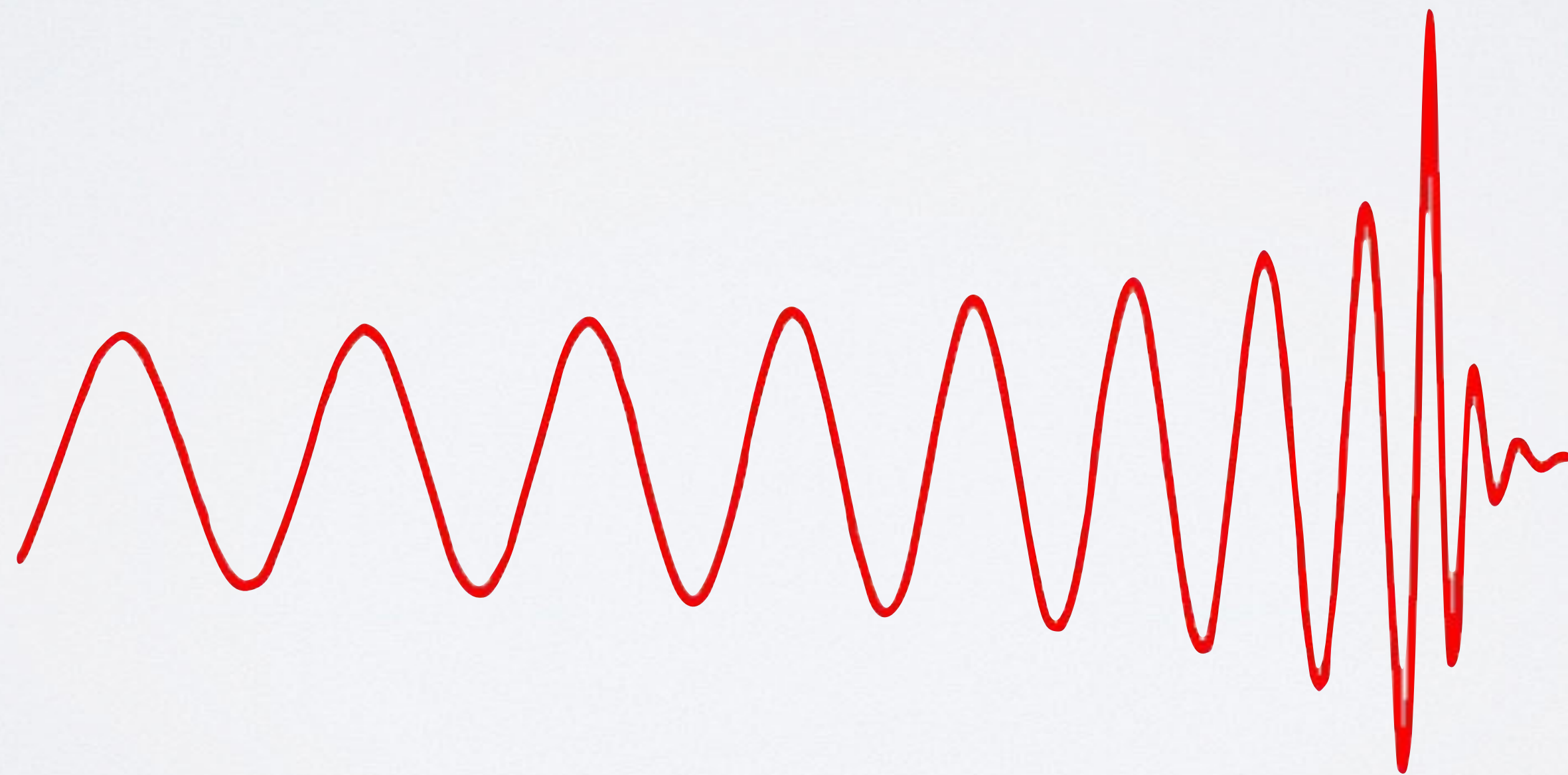
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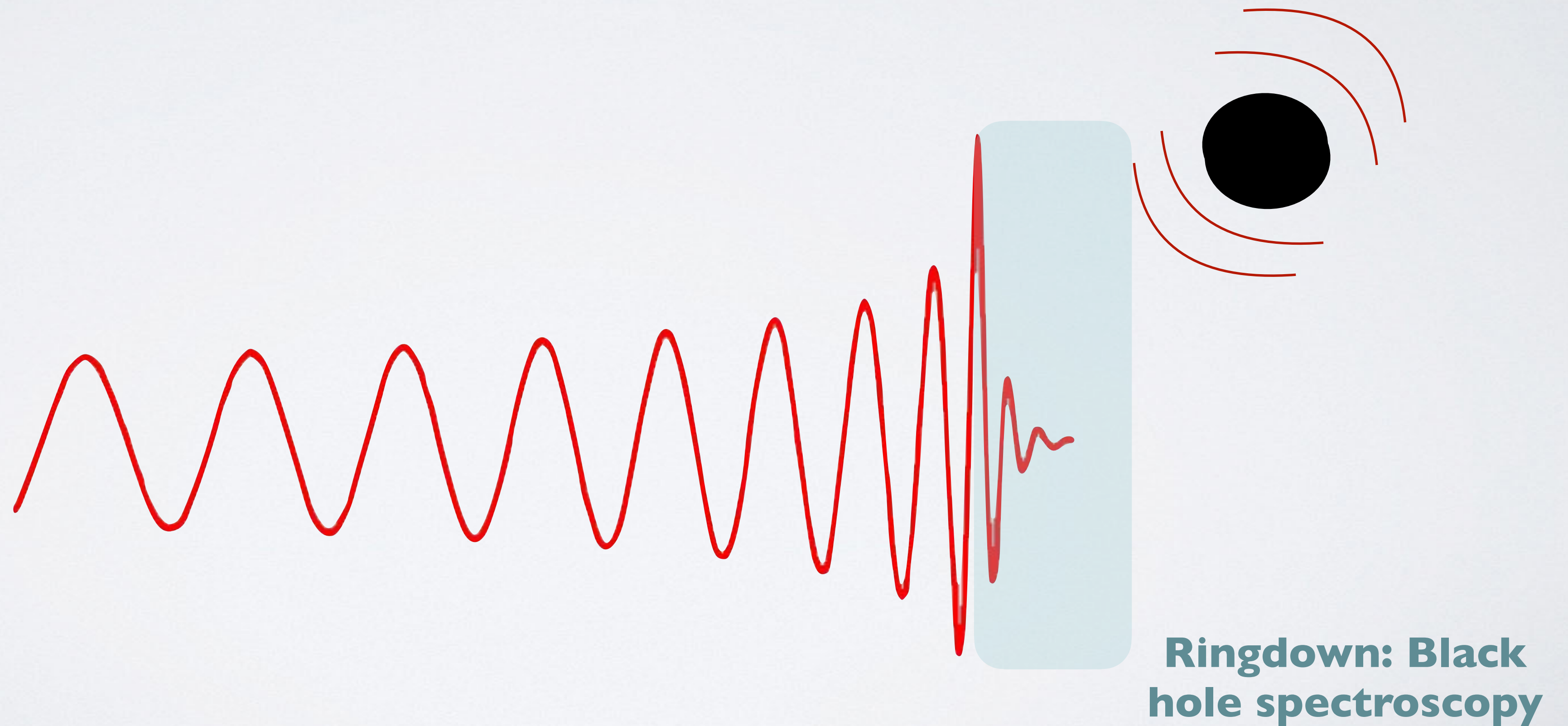
BLACK HOLE PERTURBATION THEORY

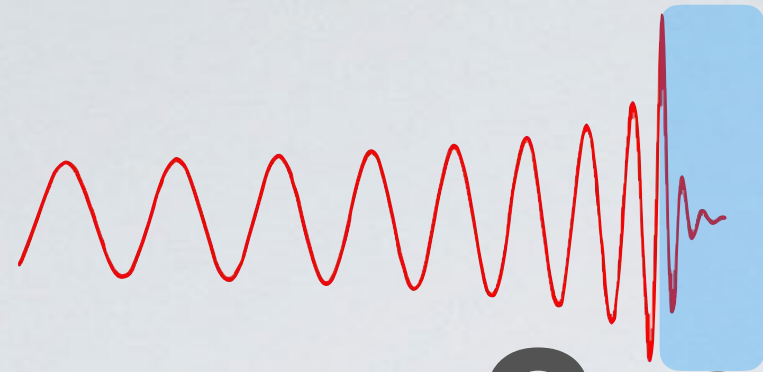
$$g_{ab} = g_{ab}^o + \epsilon h_{ab}^{(1)} + \epsilon^2 h_{ab}^{(2)} + \mathcal{O}(\epsilon^3)$$



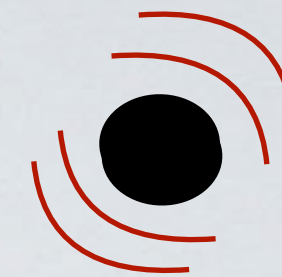
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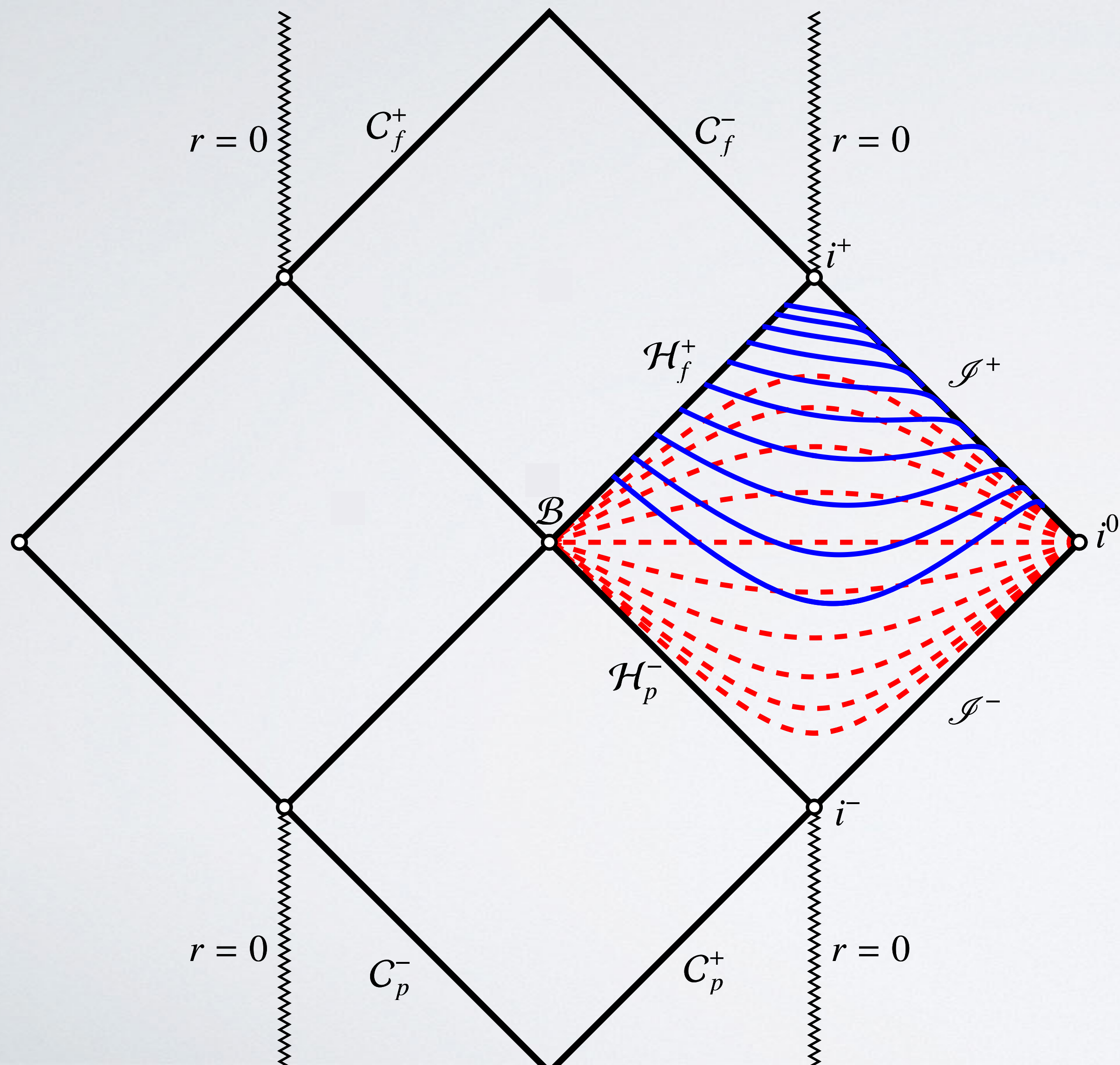


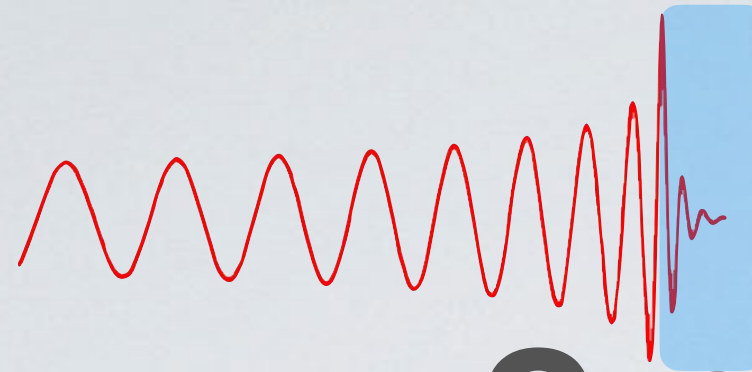


QNMS

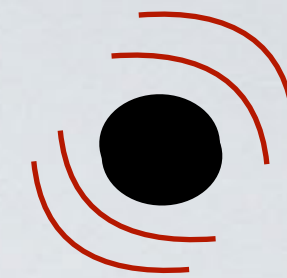


Quasinormal modes and the ringdown

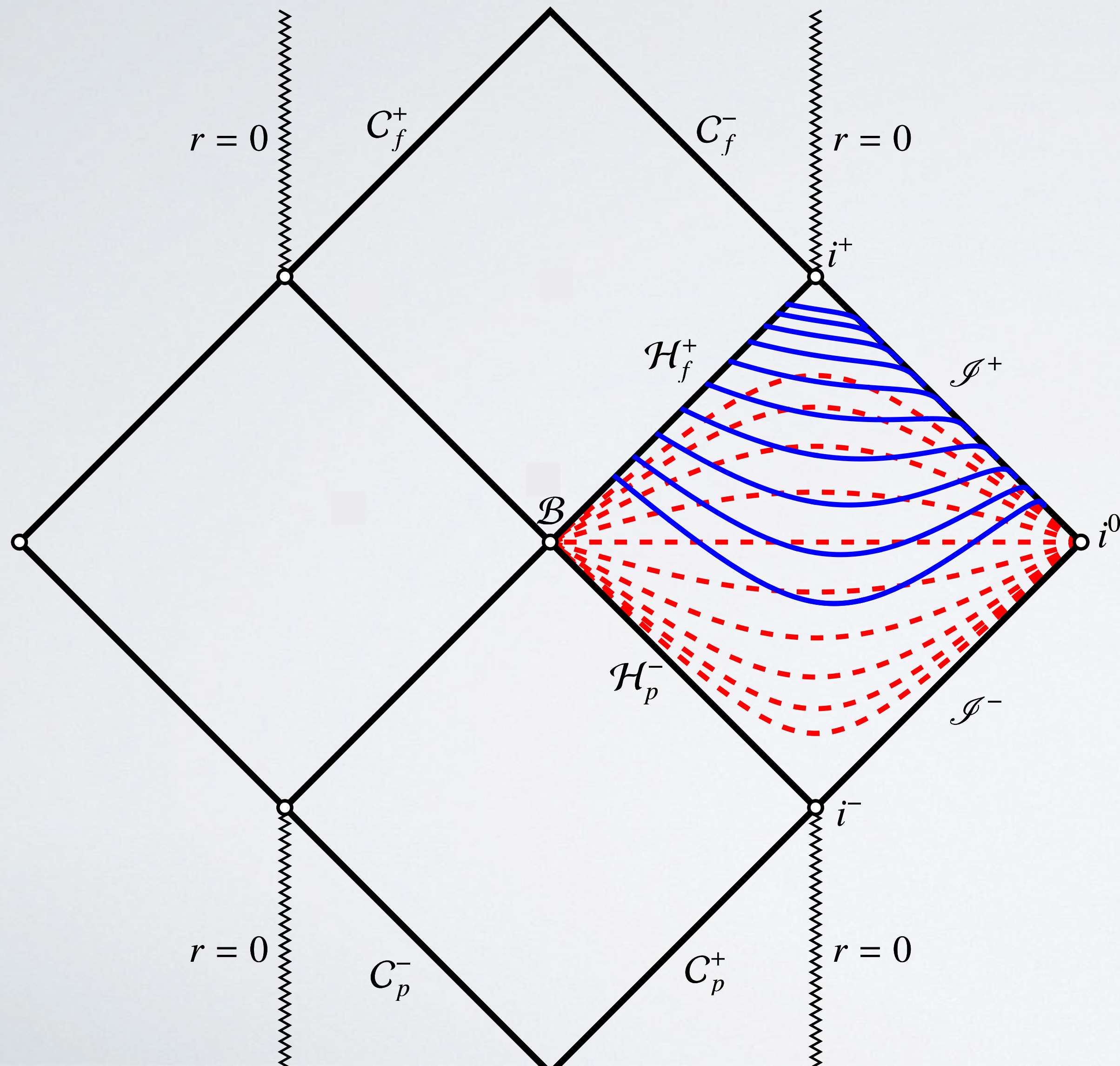




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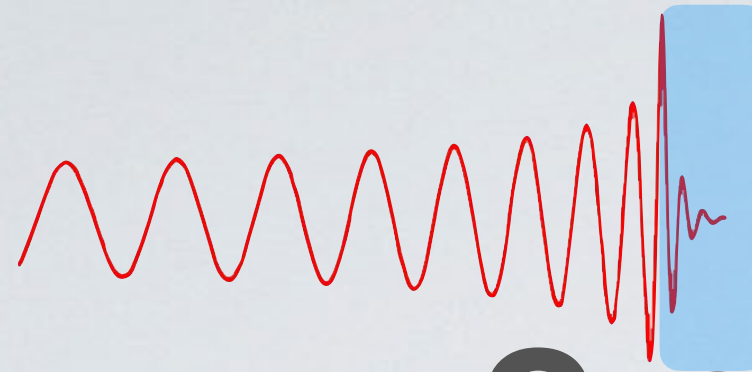


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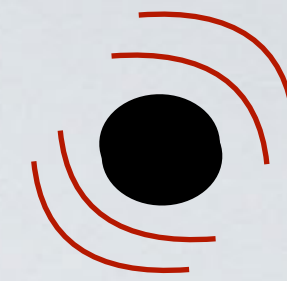


QNMs mode modelling:

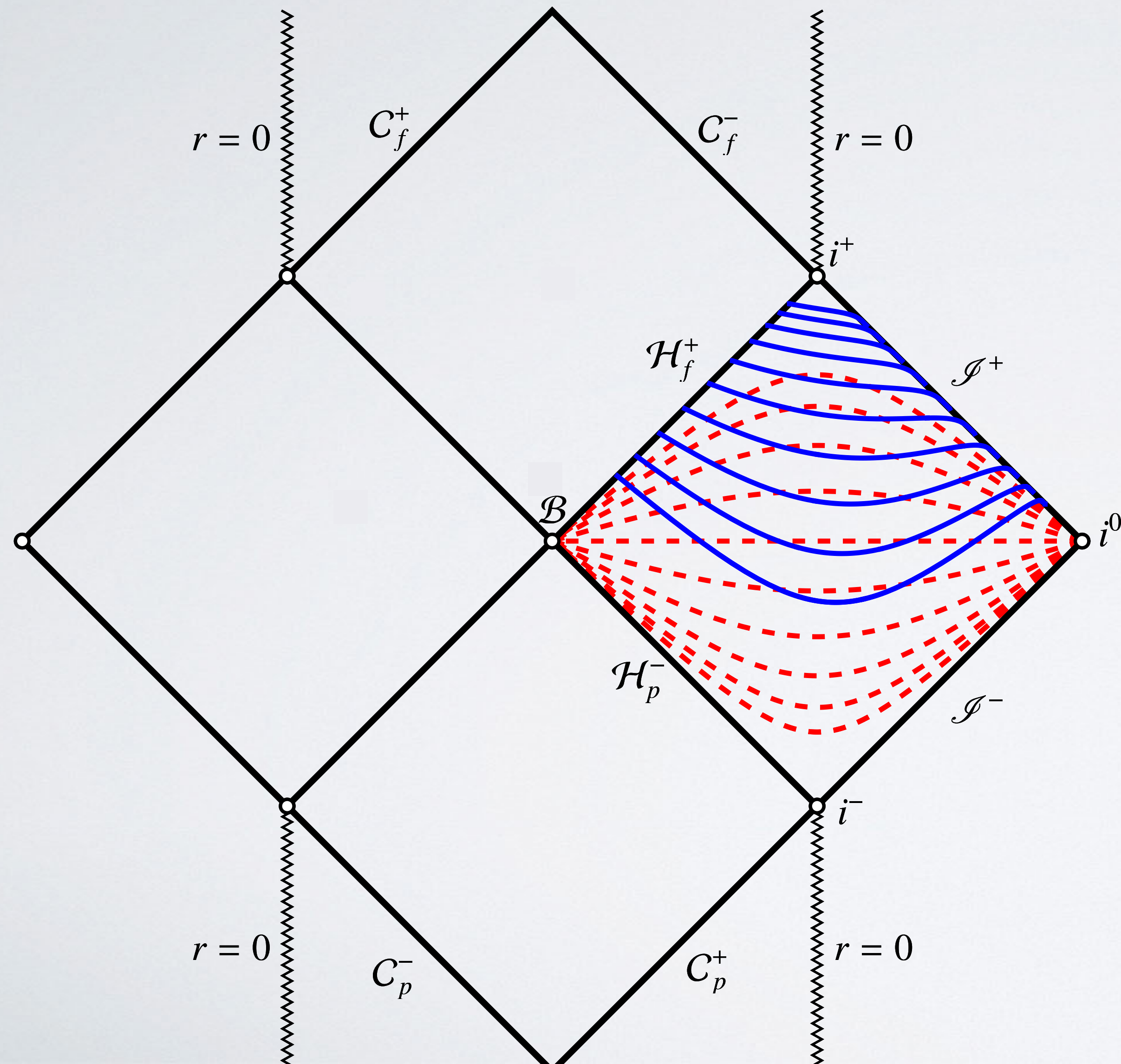
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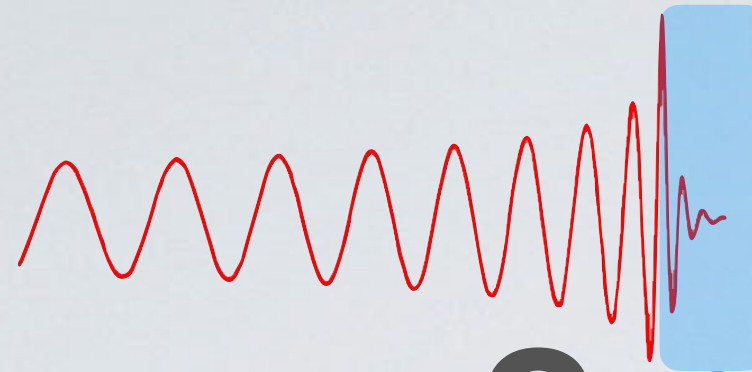


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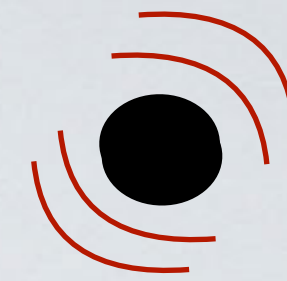
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Boundary Conditions

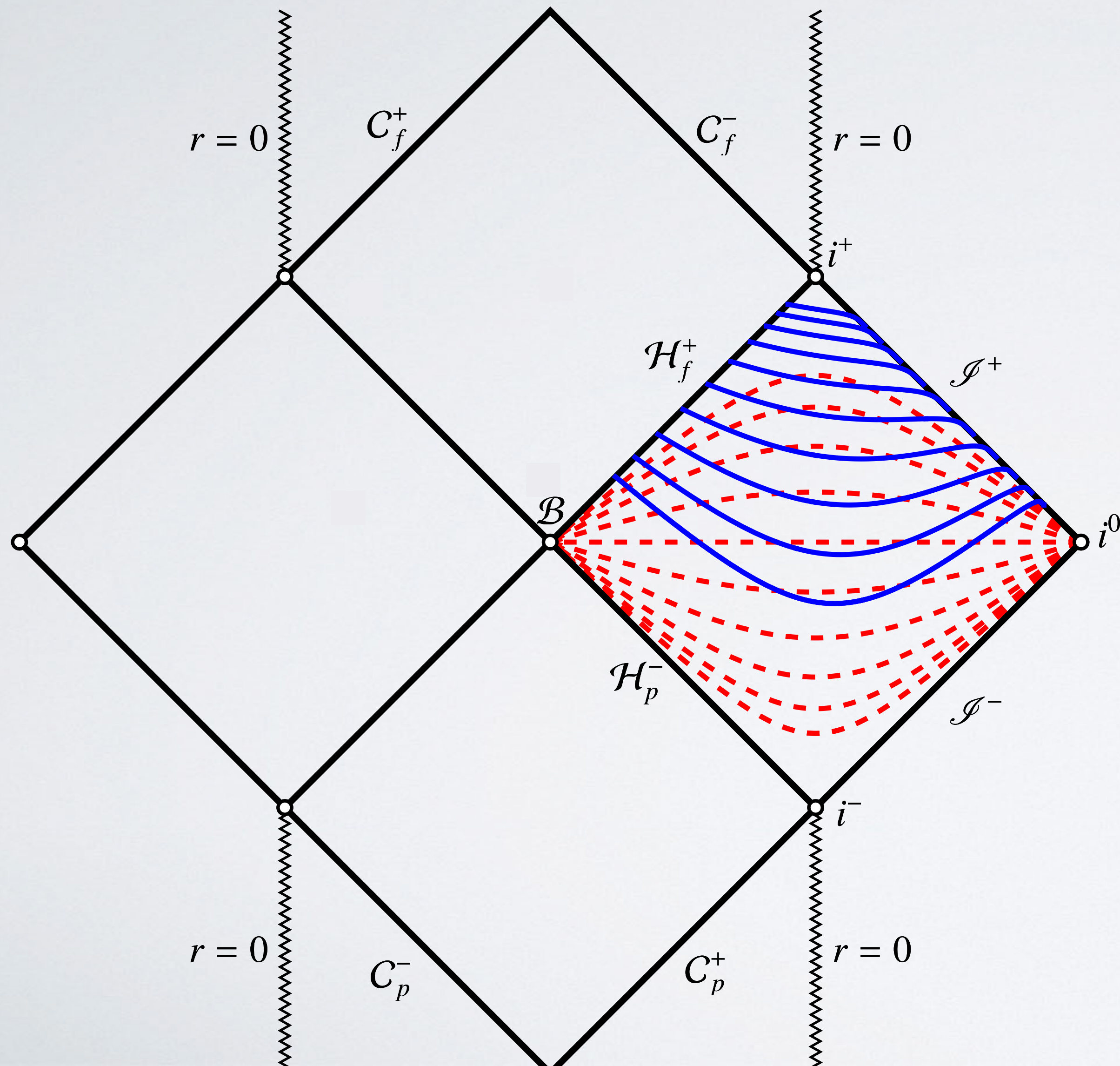
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QNMS



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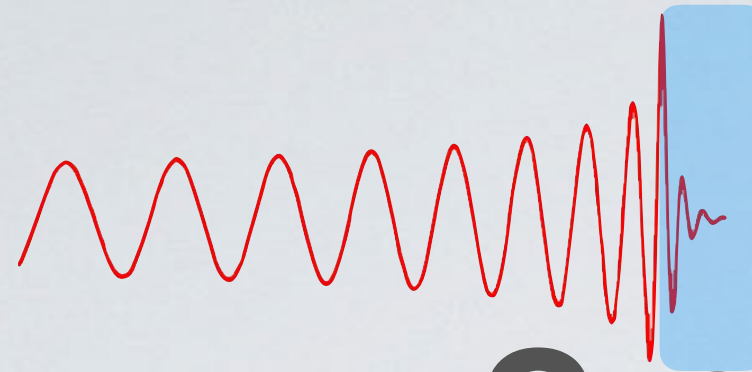
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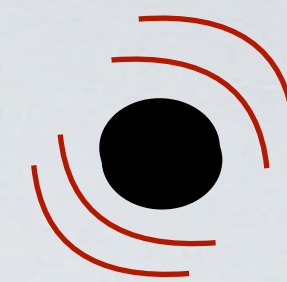
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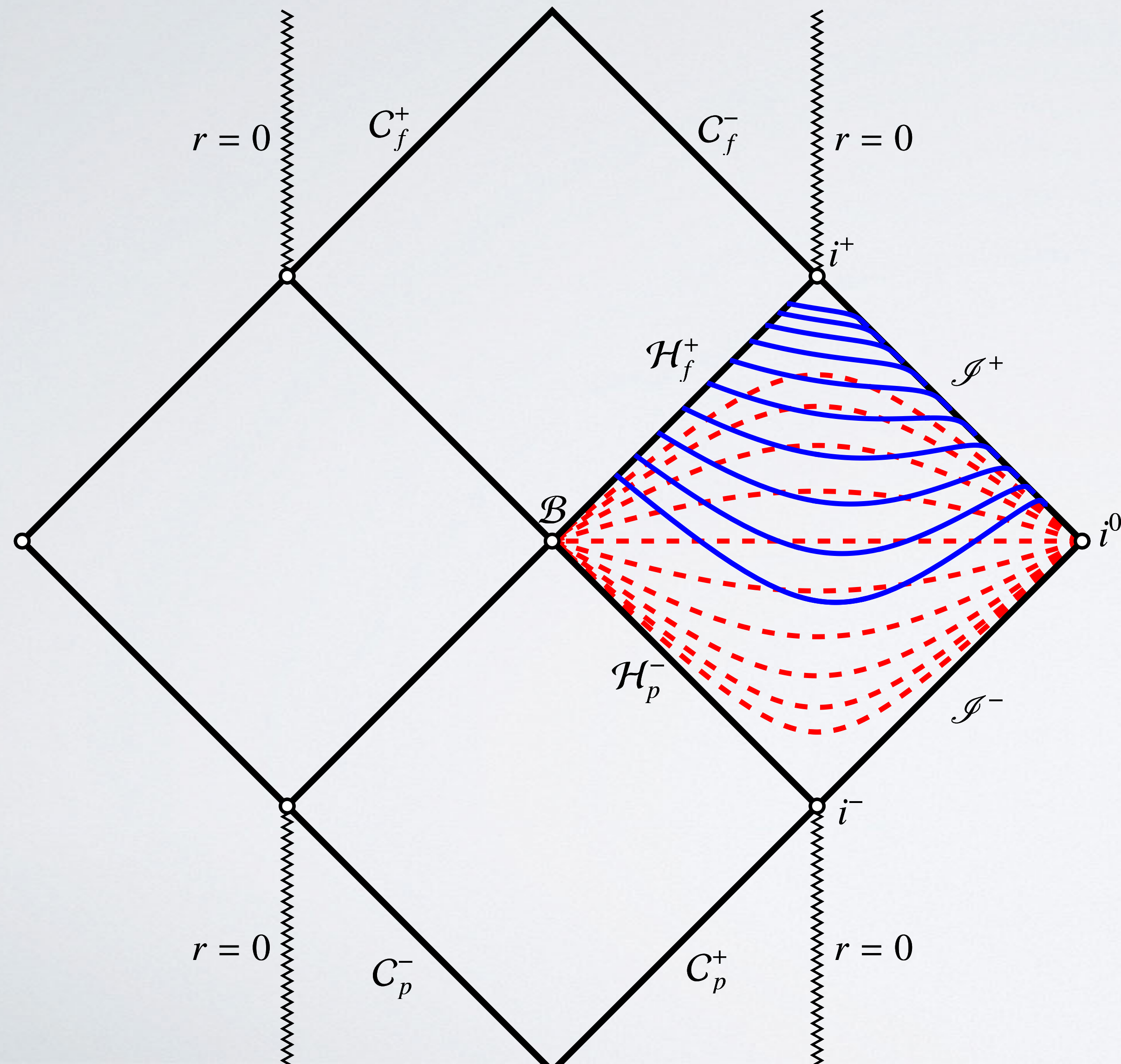
$$A_{n\ell m}(r_*) \rightarrow \infty, \quad r_* \rightarrow \pm\infty$$



QNMS



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QNM mode modelling:

$$\chi(\tau, \sigma) = \sum_n \sum_{\ell, m} \bar{A}_{n\ell m}(\sigma) e^{-i\omega_{n\ell m}\tau}$$

Regularity Conditions

$$||\bar{A}_{n\ell m}(\sigma)|| < \infty$$

QNM ORTHOGONALITY RELATIONS

— Construct products such that $\langle \Phi_{n'} | \Phi_n \rangle \propto \delta_{n'n}$

“BH Spectroscopy: from theory to experiment” (Chapter 2.5) - ArXiv 2505.23895

[A. Zimmerman + (2015) ; S. Green + (2022); L. London (2023) ; S. Ma, H. Yang (2024), P. Arnaudo (2025)]

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Analytic continuation: Deform the *physical* radial coordinate into the complex plane

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But QNM orthogonality/projections are available within the hyperboloidal framework without the need of any regularisation strategy

[M. Ansorg, RPM (2016) ; RPM, J.L. Jaramillo, M. Ansorg (2018);
E. Gasperin, J.L. Jaramillo (2022) ; J. Besson, J.L. Jaramillo (2025)]

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$t \rightarrow -t, \phi \rightarrow -\phi$ SYMMETRY

(i) Given a solution $\Phi(t, r, \theta, \phi)$ to the wave equation

(ii) Operator $\mathcal{J}$ act as $\mathcal{J}\Phi = \Phi|_{t \rightarrow -t, \phi \rightarrow -\phi}$

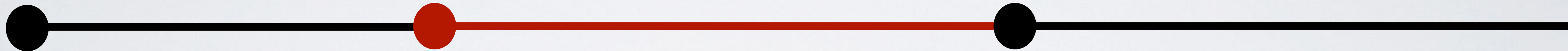
(ii) Orthogonality product $\langle \Phi_{n'} | \Phi_n \rangle \sim \{ \mathcal{J}\Phi_{n'} \} \cdot \{ \Phi_n \}$

OUTLINE

QNM products

$t-\phi$ 'Symmetry'

Future/past hyperboloidal



KERR SPACETIME

— Boyer-Lindquist coordinates $x^\mu = \{t, r, \theta, \phi\}$

$$ds^2 = -f(r, \theta)dt^2 - \frac{4Mar}{\Sigma(r, \theta)} \sin^2 \theta dt d\phi + \frac{\Sigma(r, \theta)}{\Delta(r)} dr^2 + \Sigma(r, \theta) d\theta^2 \\ + \sin^2 \theta \left(r^2 + a^2 + \frac{2Ma^2r}{\Sigma(r, \theta)} \sin^2 \theta \right) d\phi^2$$

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WAVE EQUATION $\square\Phi = 0$

— Boyer-Lindquist coordinates $x^\mu = \{t, r, \theta, \phi\}$

$$\square := - \left(\frac{(r^2 + a^2)^2}{\Delta} - a^2(1 - x^2) \right) \partial_t^2 - \frac{4Mar}{\Delta} \partial_t \partial_\phi + \Delta \partial_r^2 + \partial_r \Delta \partial_r \\ + (1 - x^2) \partial_x^2 - 2x \partial_x - \left(\frac{a^2}{\Delta} - \frac{1}{1 - x^2} \right) \partial_\phi^2$$

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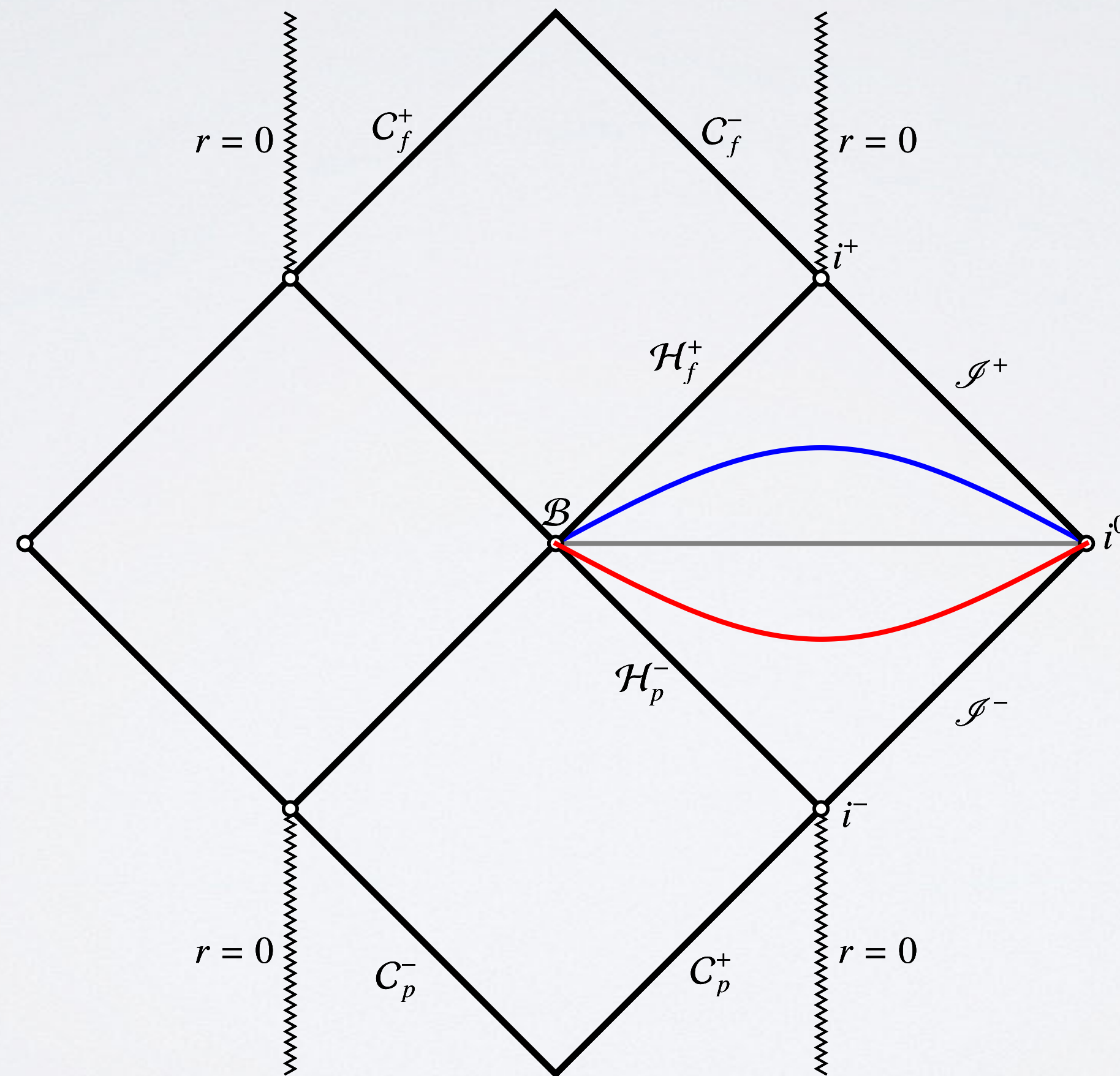
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$t-\phi$ SYMMETRY

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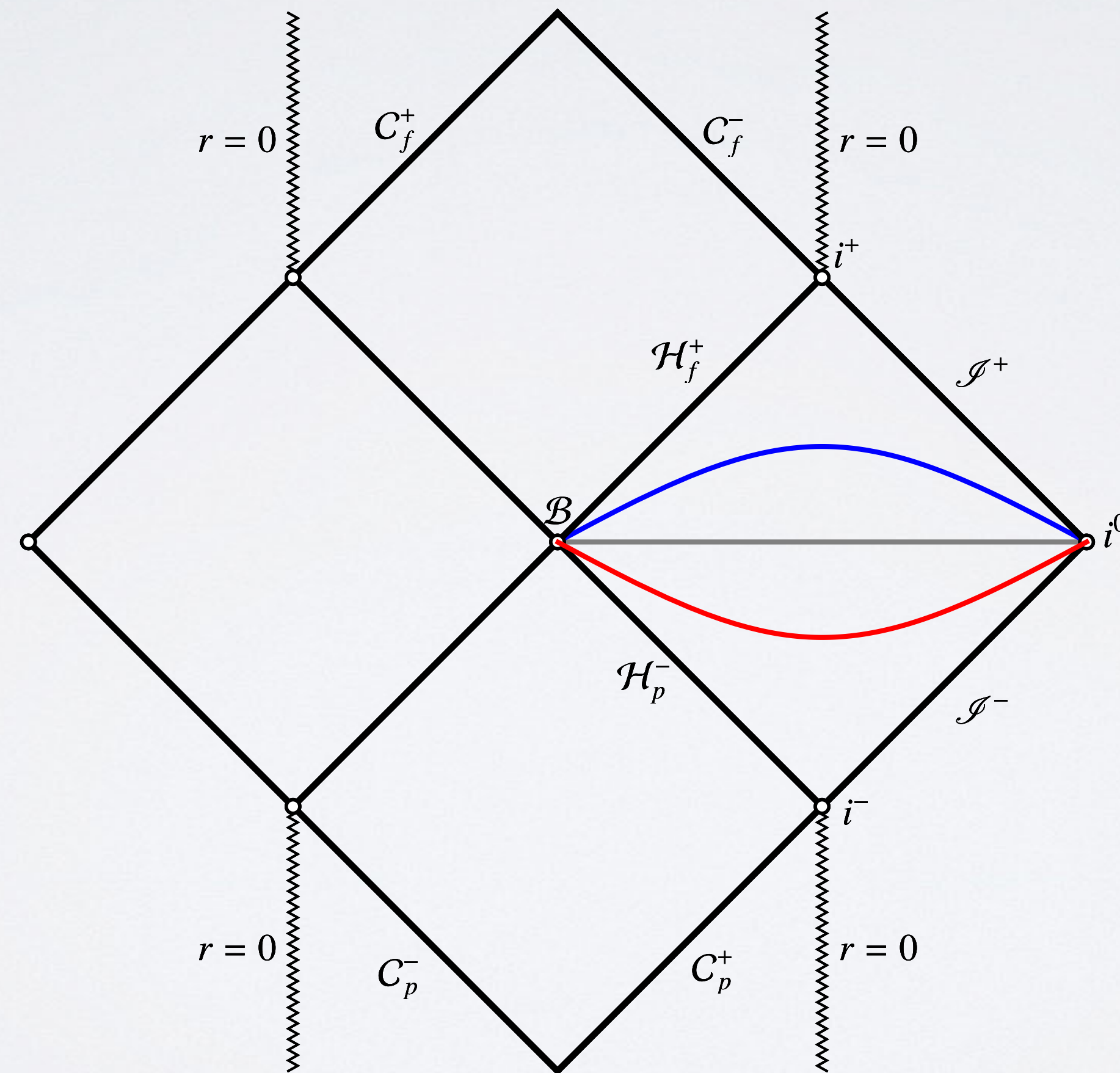
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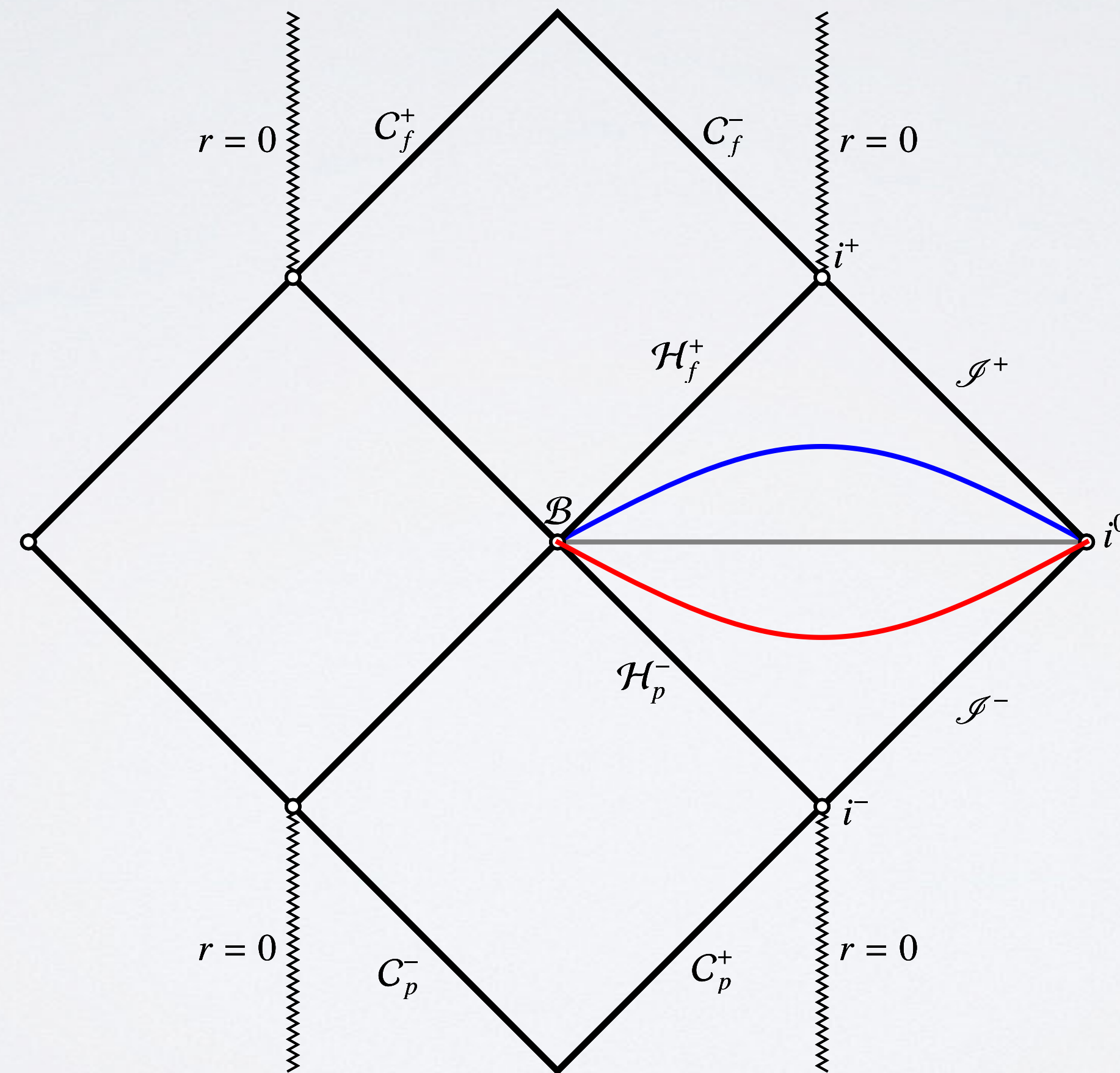
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— $t = 0$

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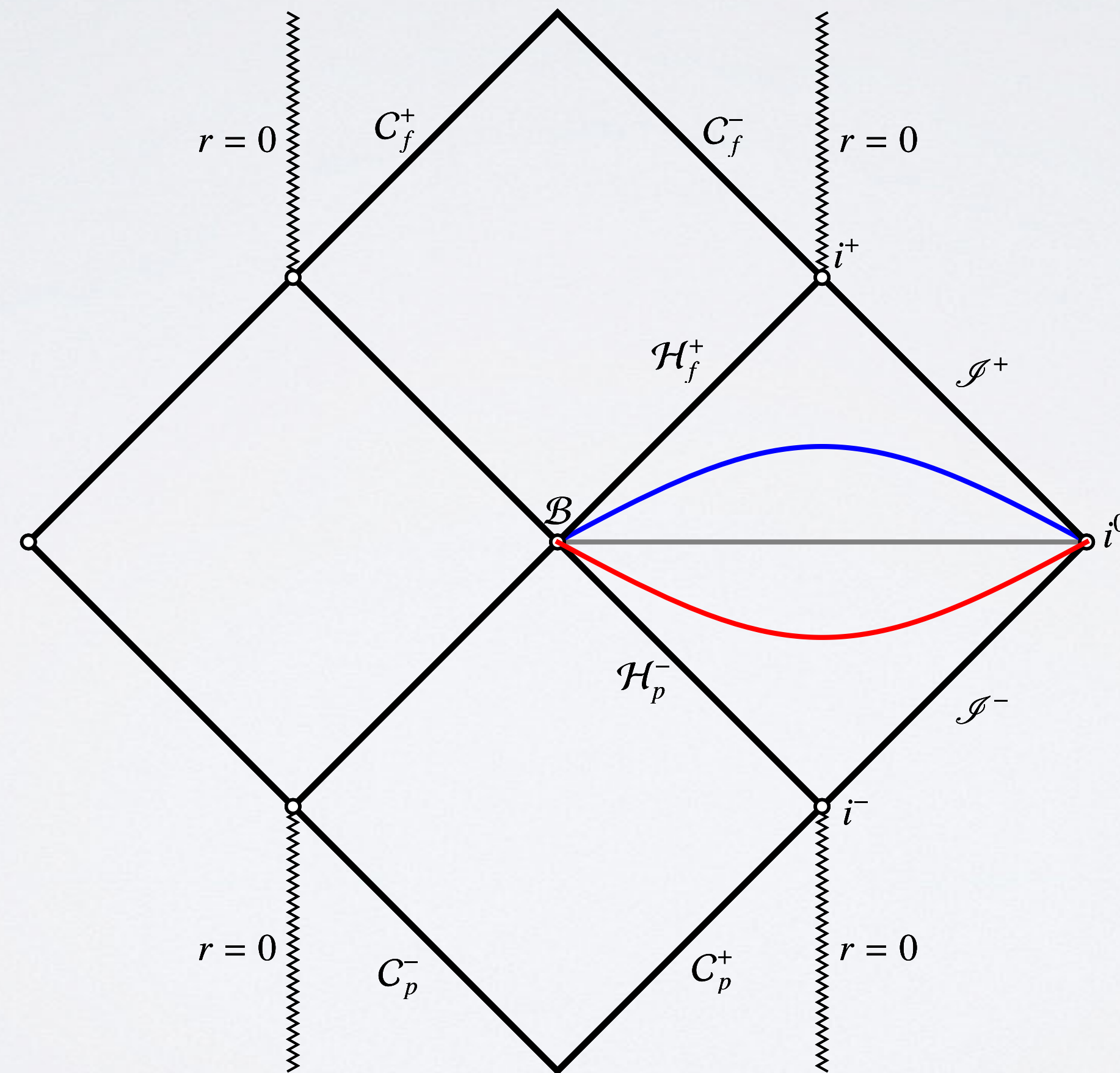


— $t = 2r_h$

— $t = 0$

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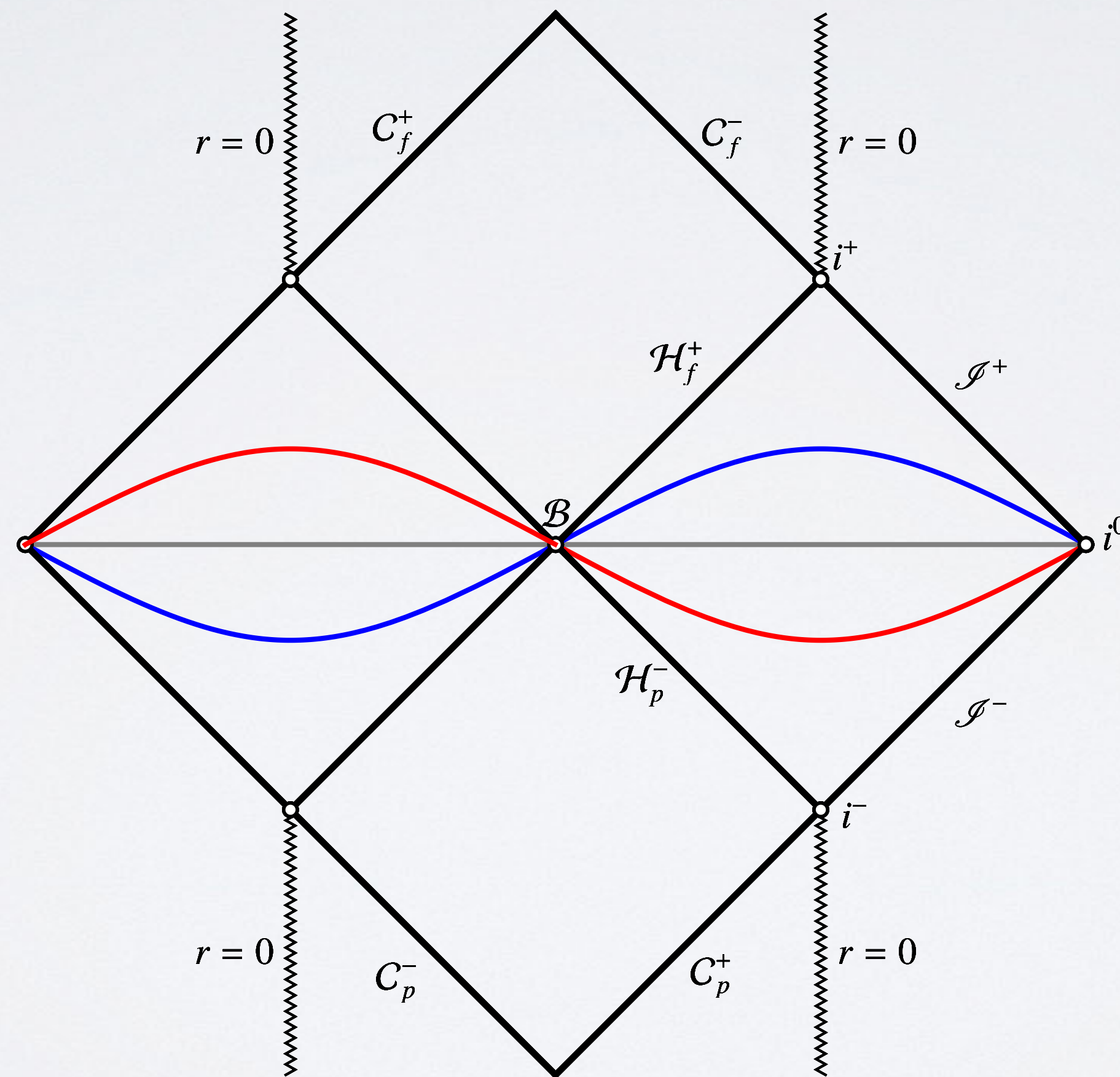
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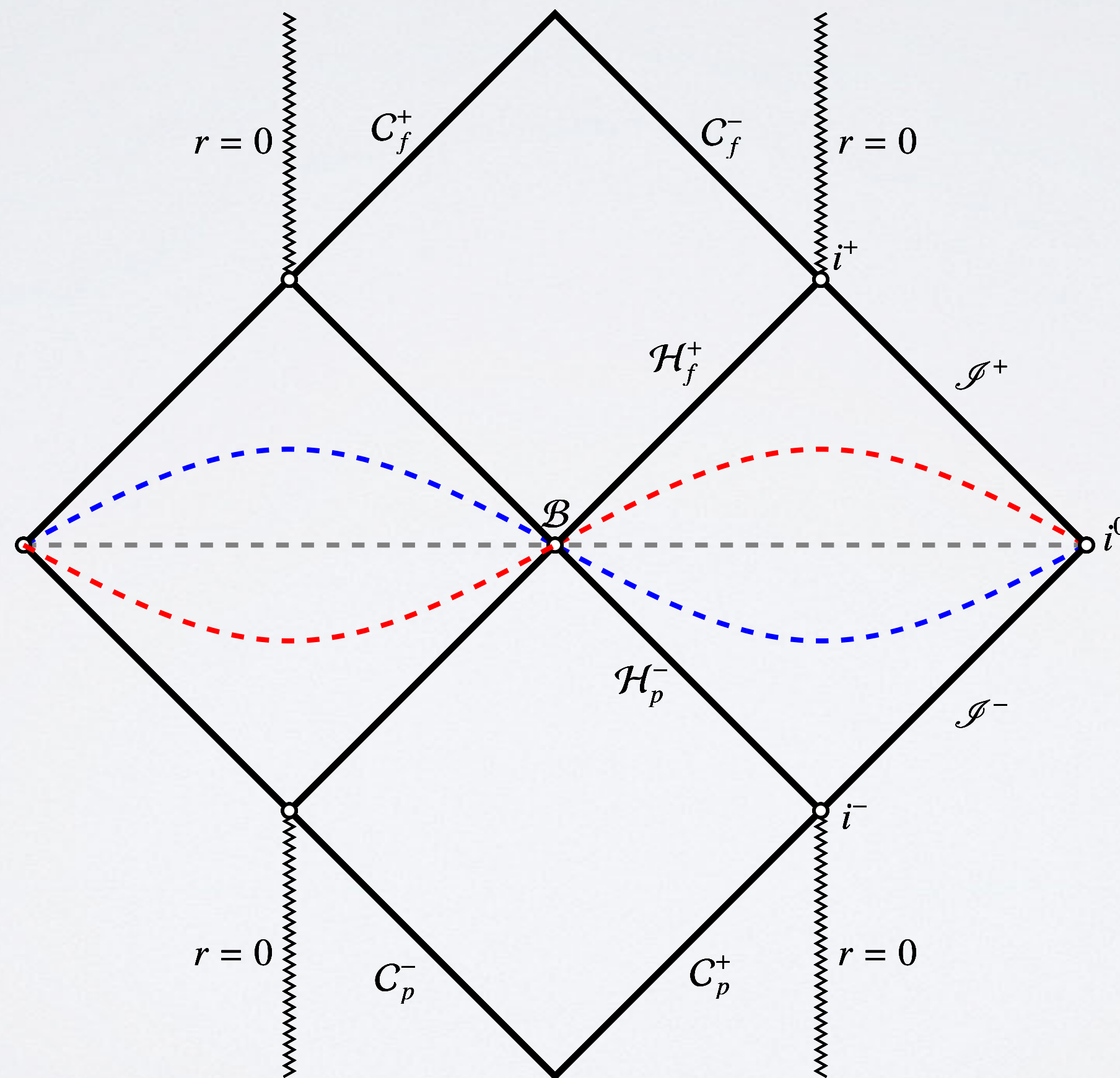
$$\phi' = -\phi$$

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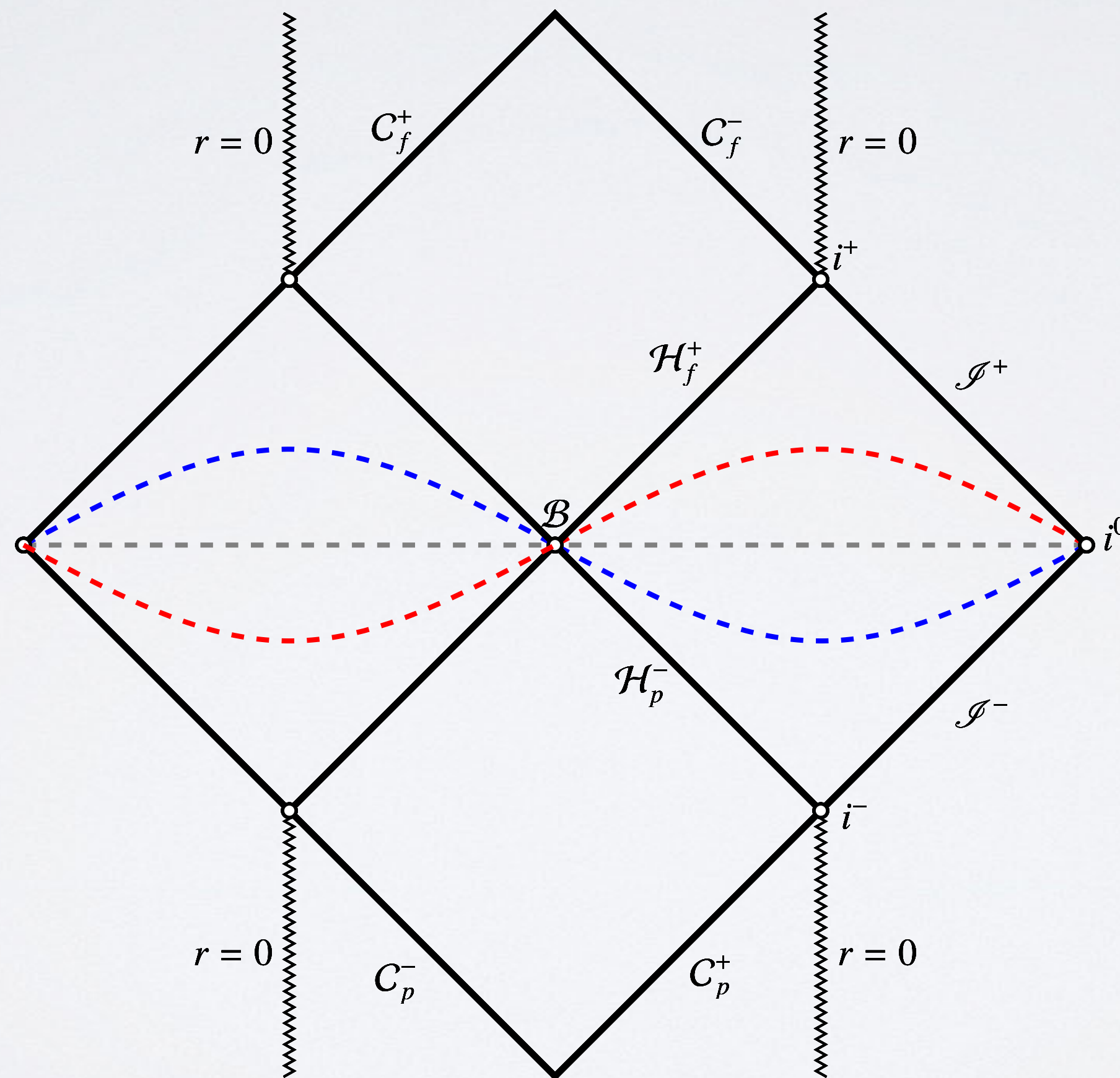


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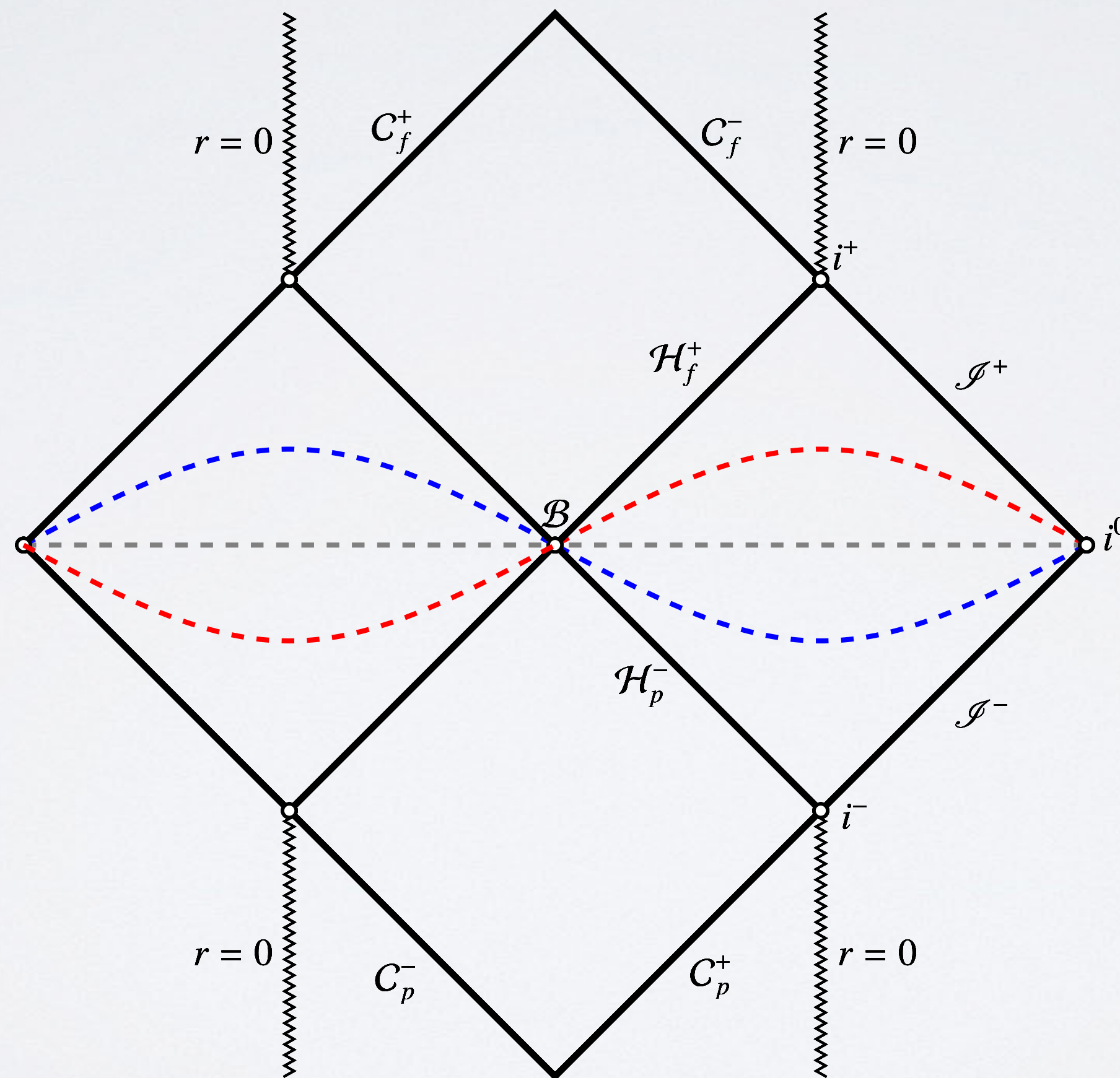


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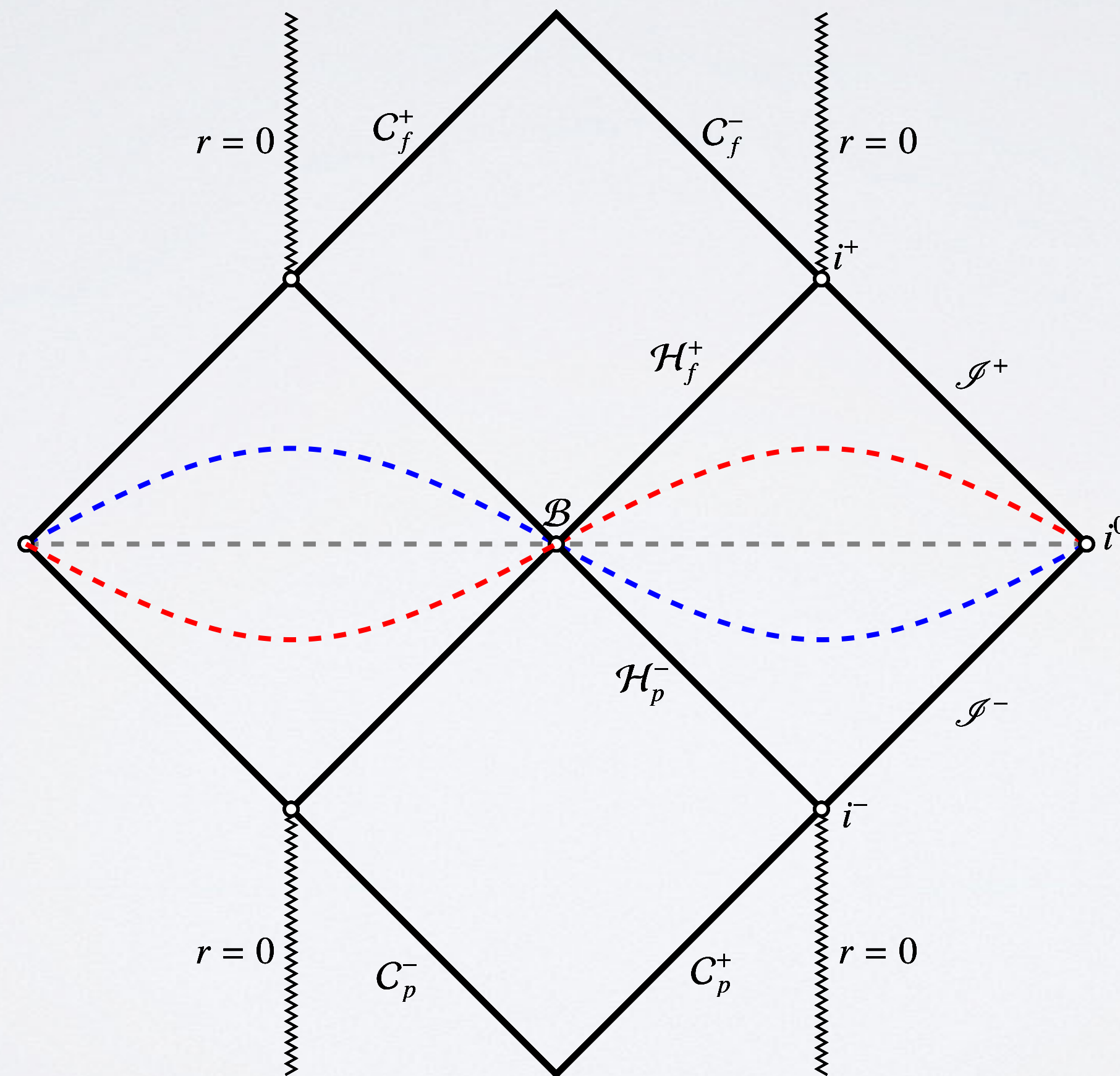
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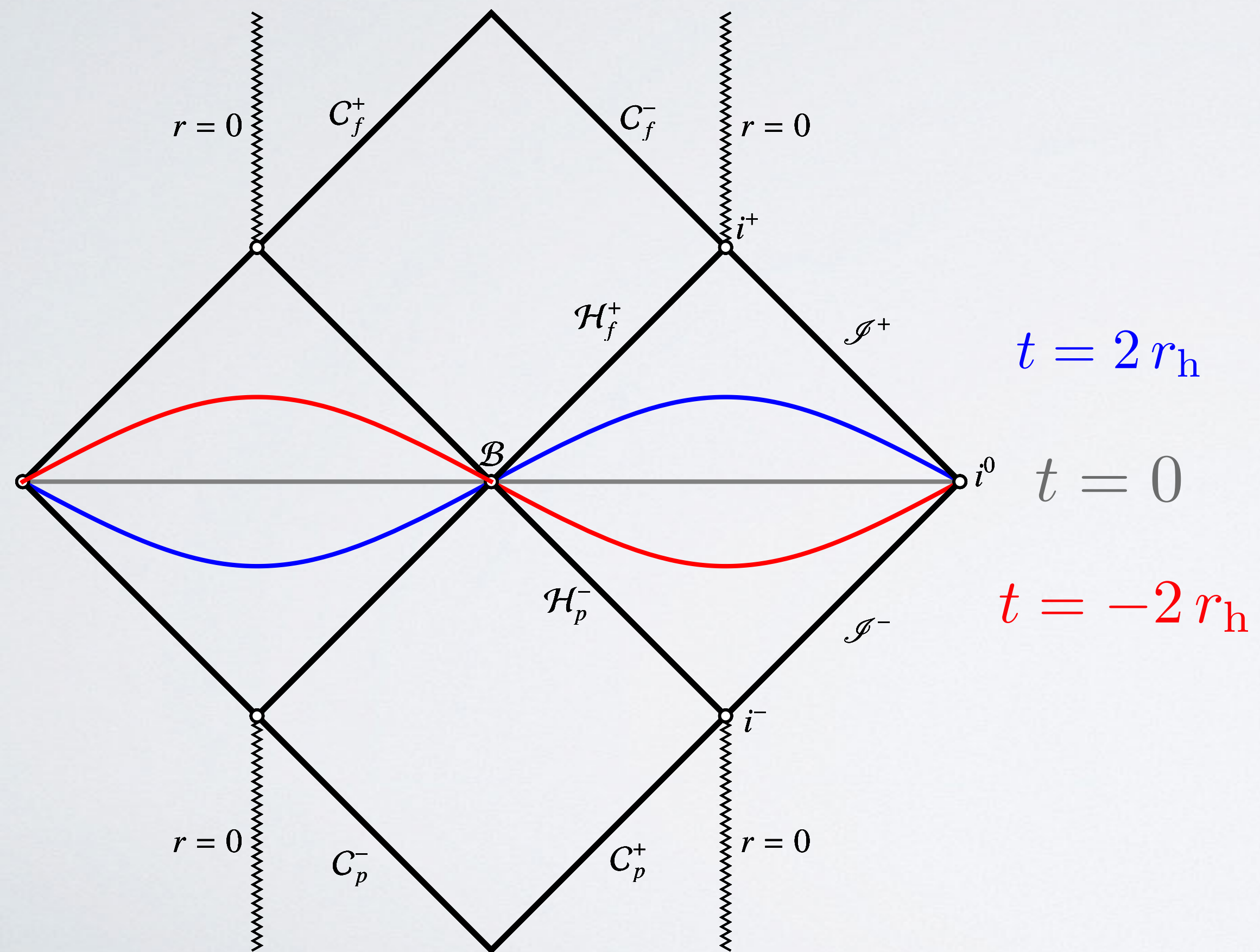
..... $t' = -2r_h$

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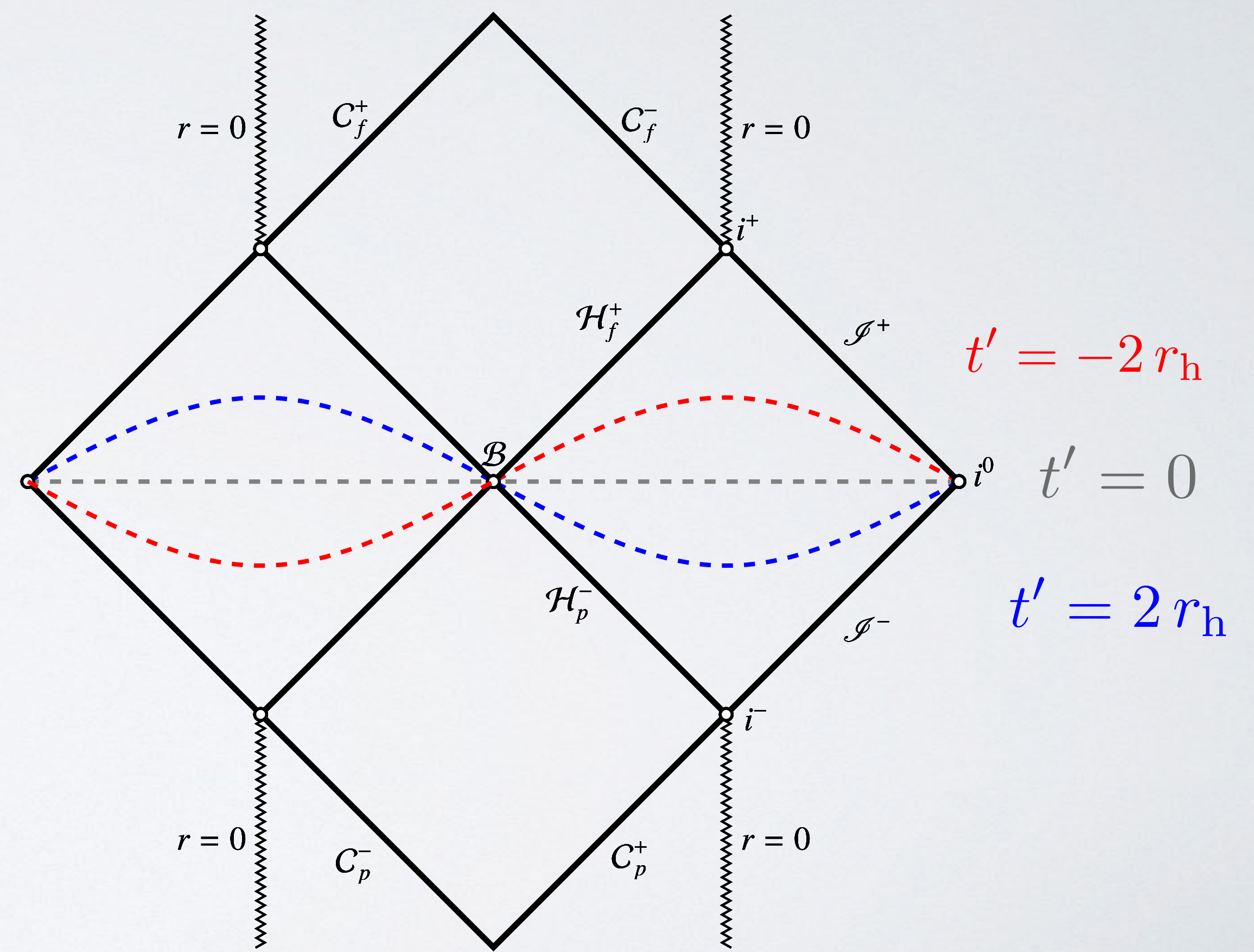
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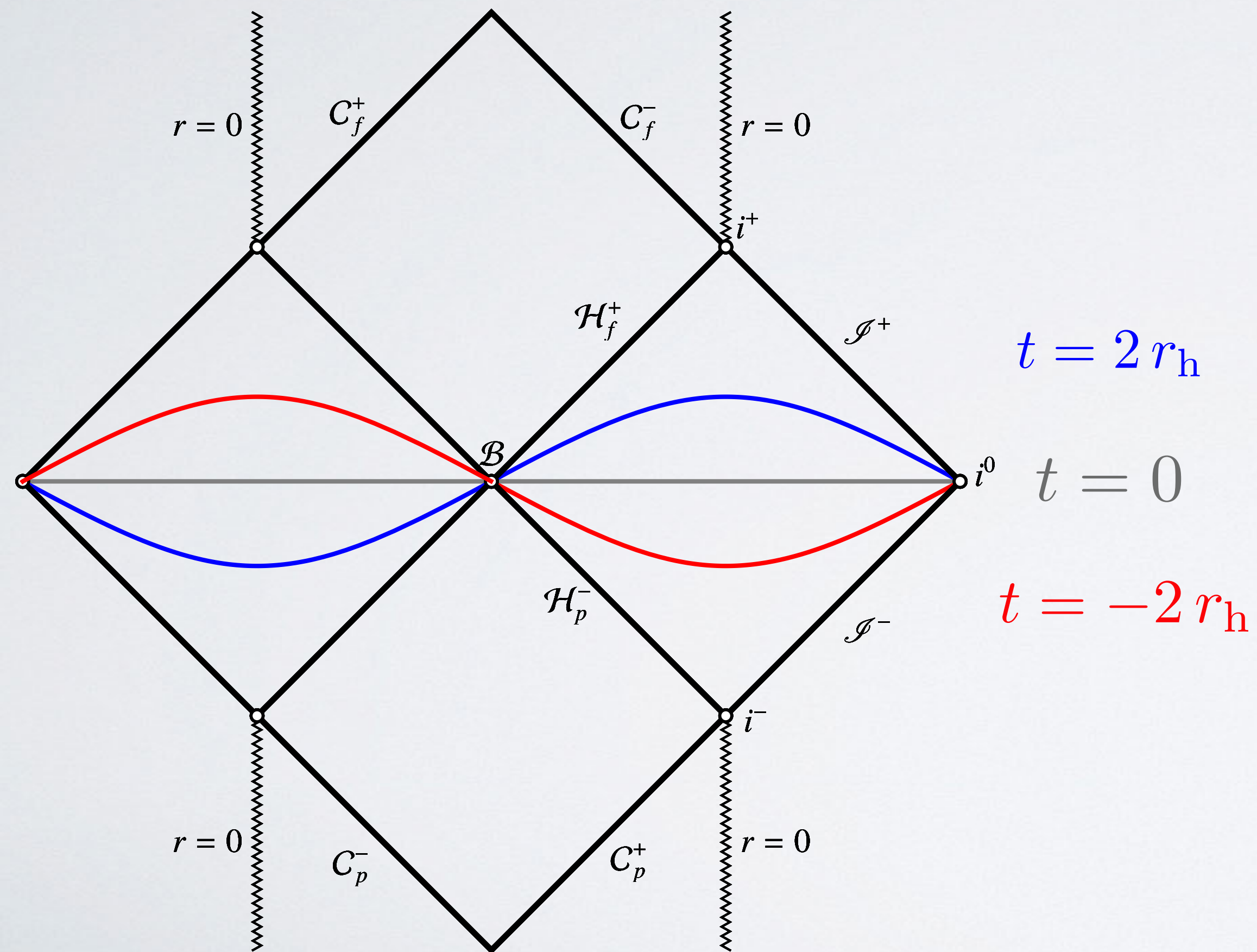


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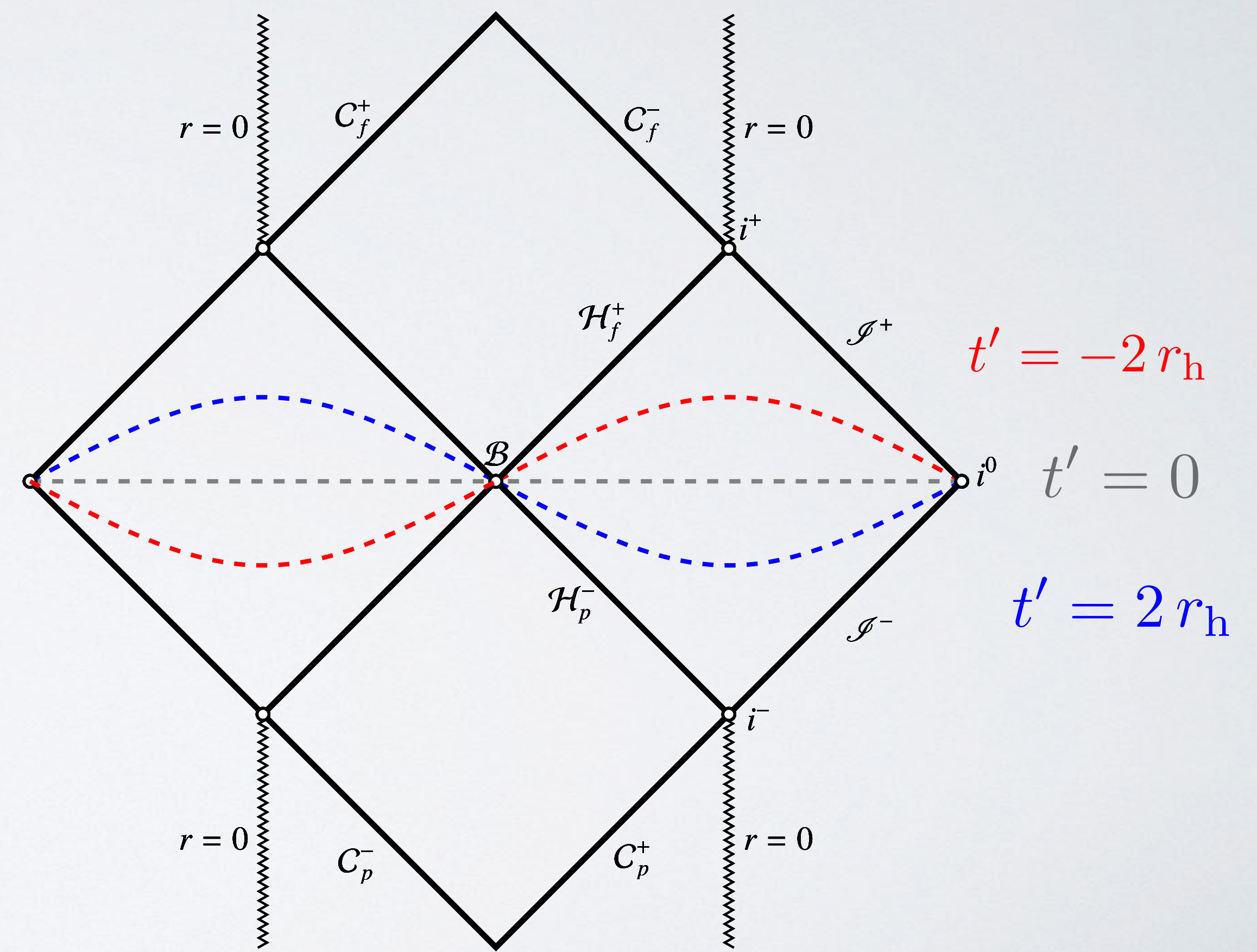


$t-\phi$ SYMMETRY !!!

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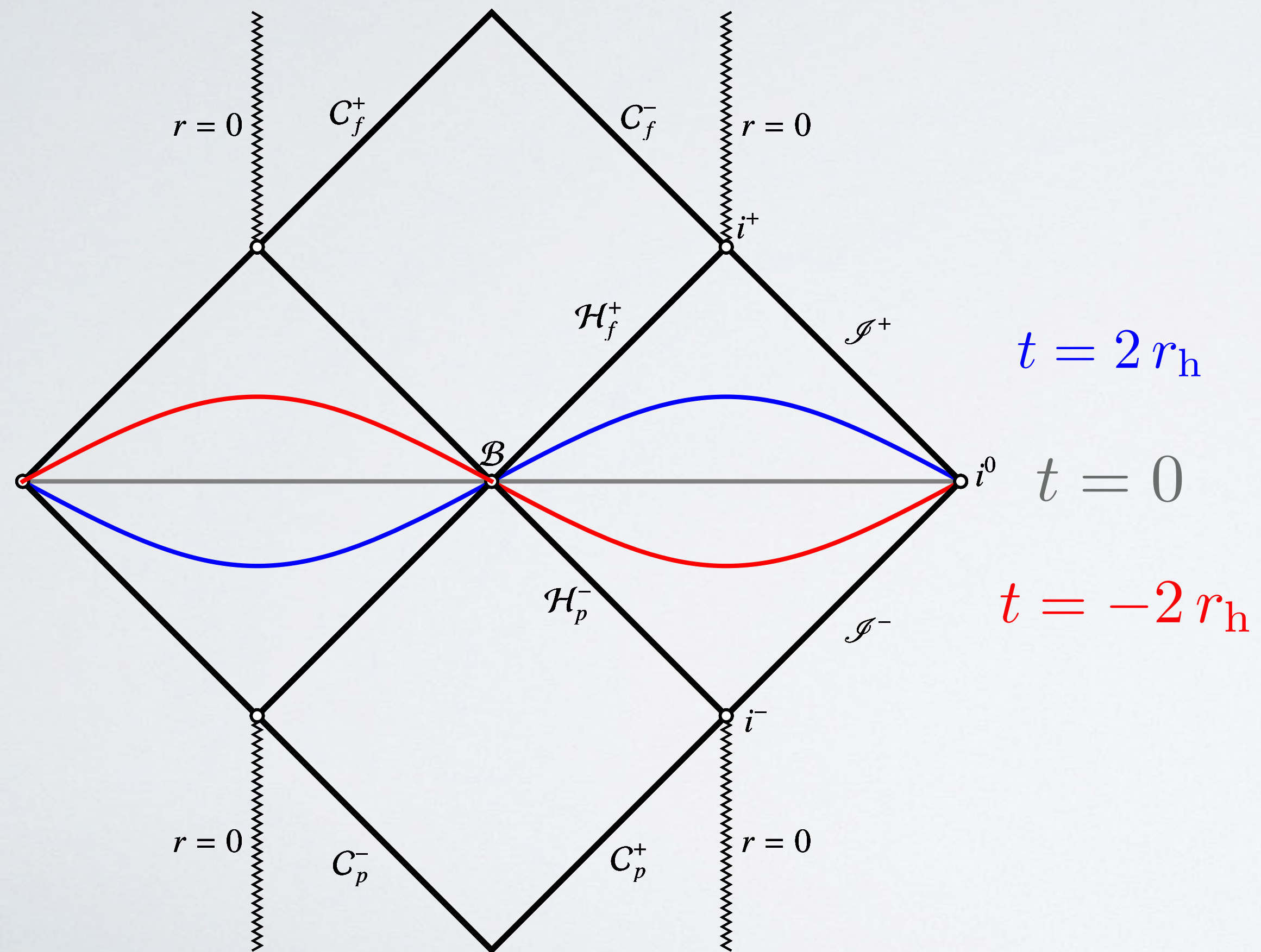


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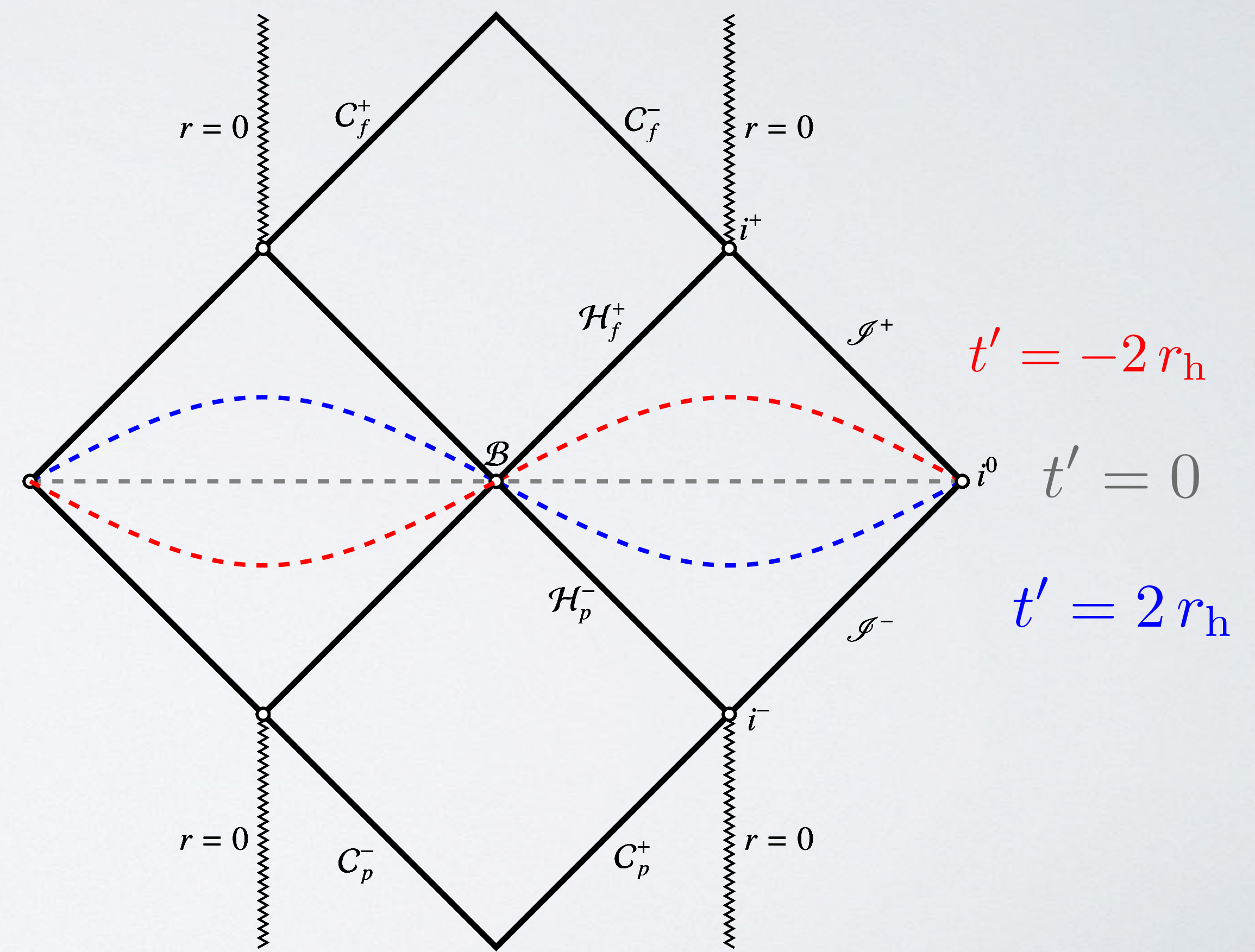


$t-\phi$ SYMMETRY !!! REALLY ?!?!?!

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WAVE EQUATION $\square\Phi = 0$

— Schwarzschild coordinates $x^\mu = \{t, r, \theta, \phi\}$

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**We have an equation.
We don't have a problem.**

[ESI - Vienna 2026]



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$$\psi_{\ell m} \sim e^{-i\omega(t \mp r_*)}$$

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$\mathcal{I}$



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$$\mathcal{J} \psi^{\text{ret}} = \psi^{\text{adv}}$$

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QNM ringdown problem

$\mathcal{J}$

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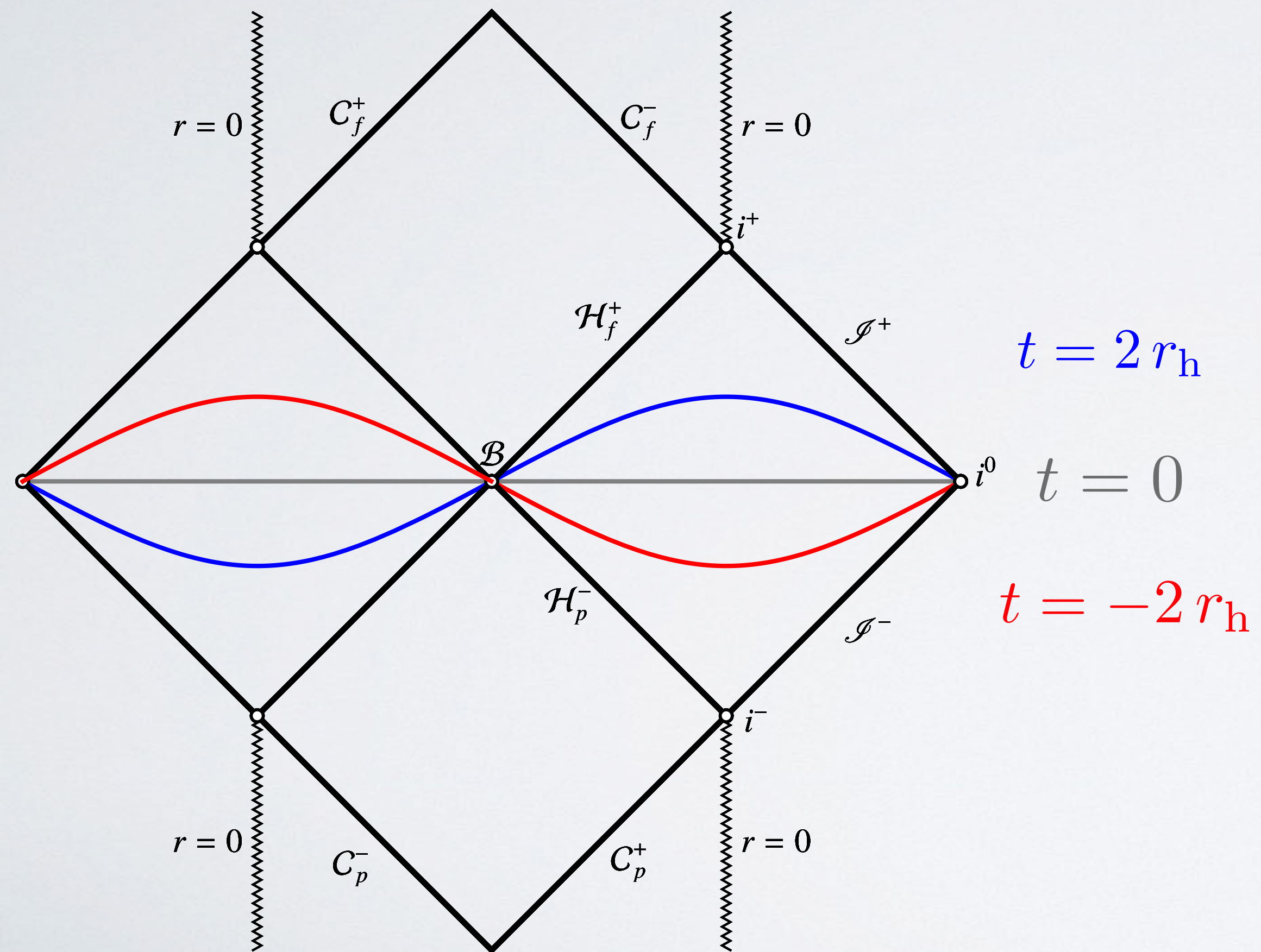
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'Anti'-QNM 'ringup' problem

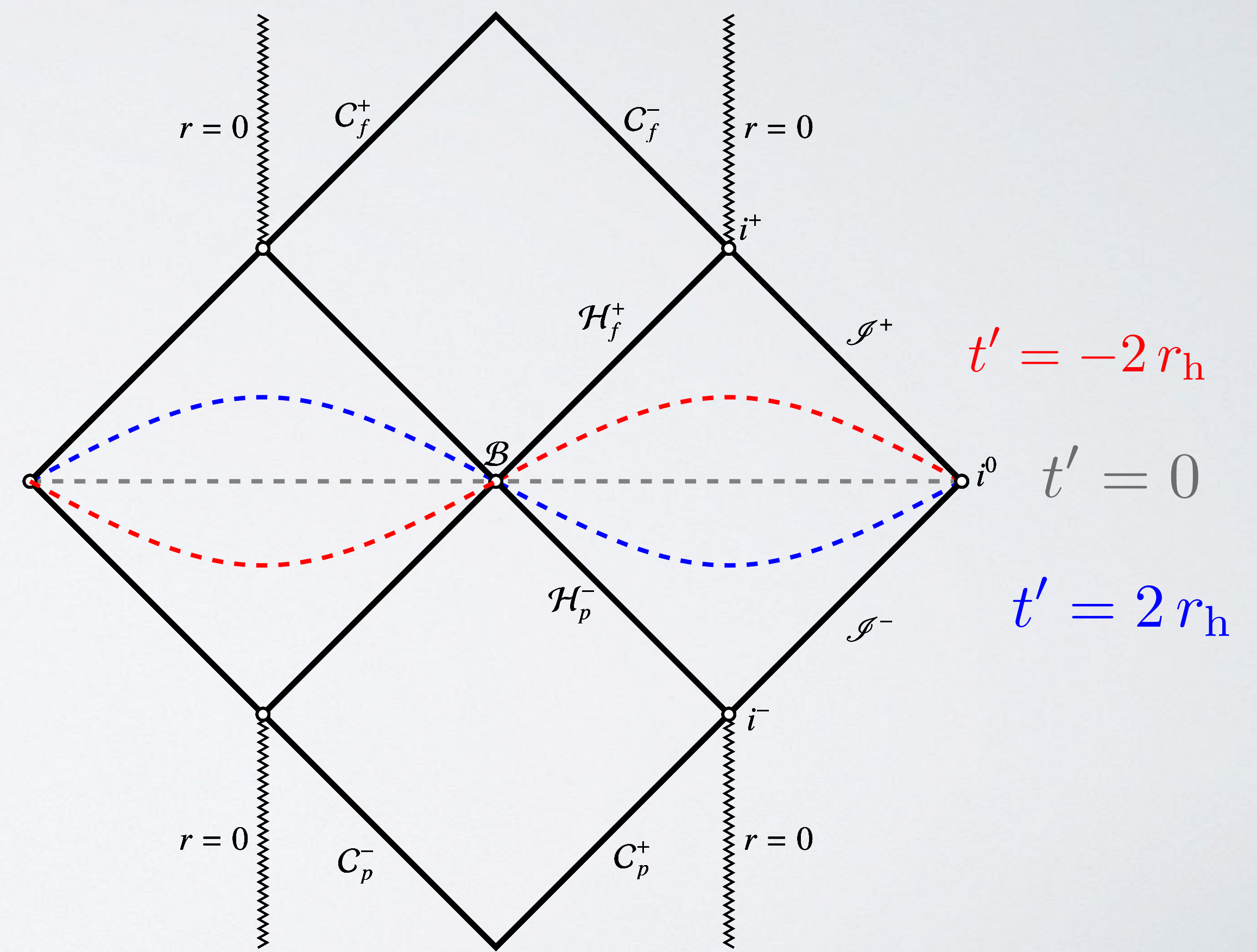
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$t-\phi$ SYMMETRY !!! REALLY ?!?!?!

$$x^\mu = \{t, r, \theta, \phi\}$$

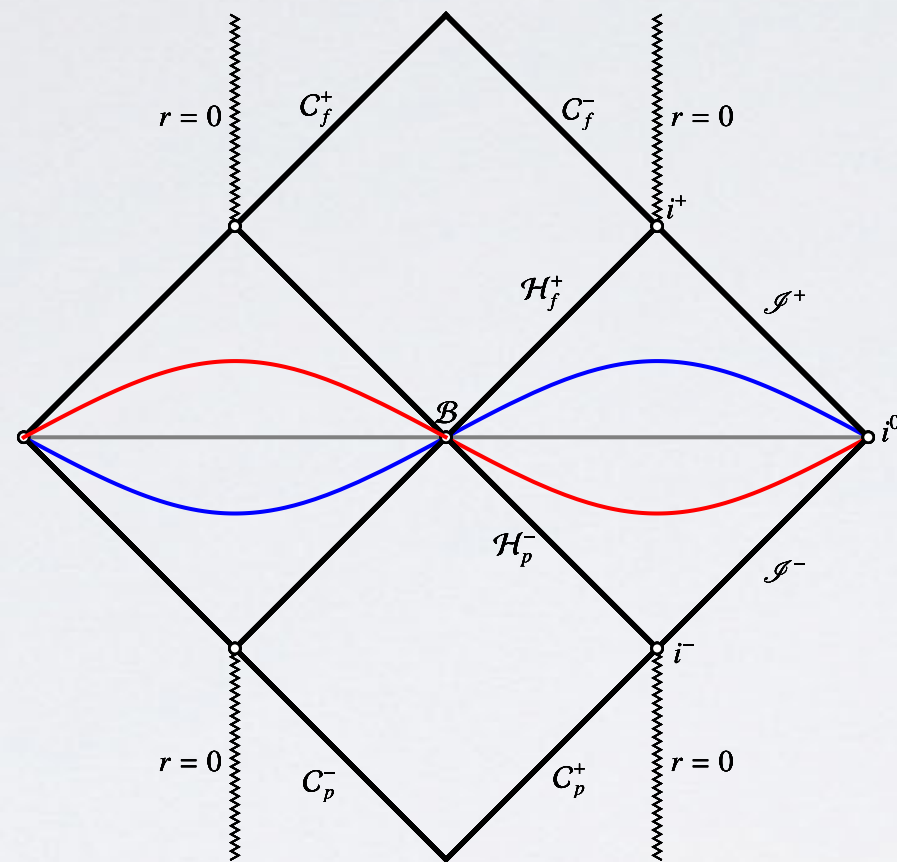


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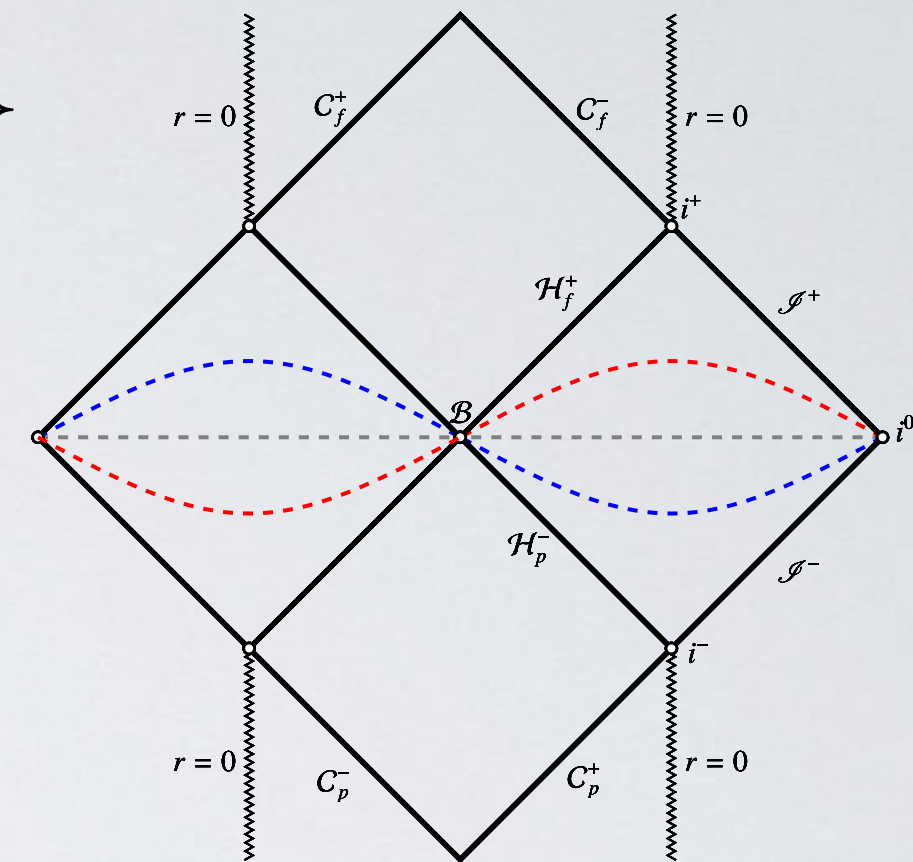


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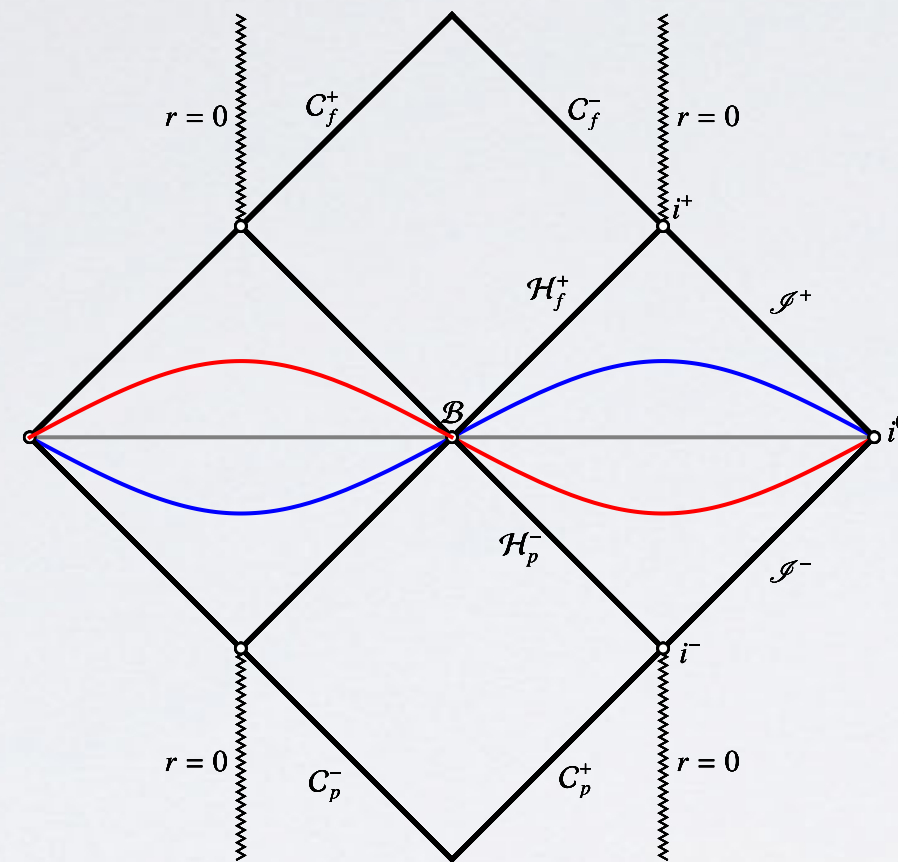


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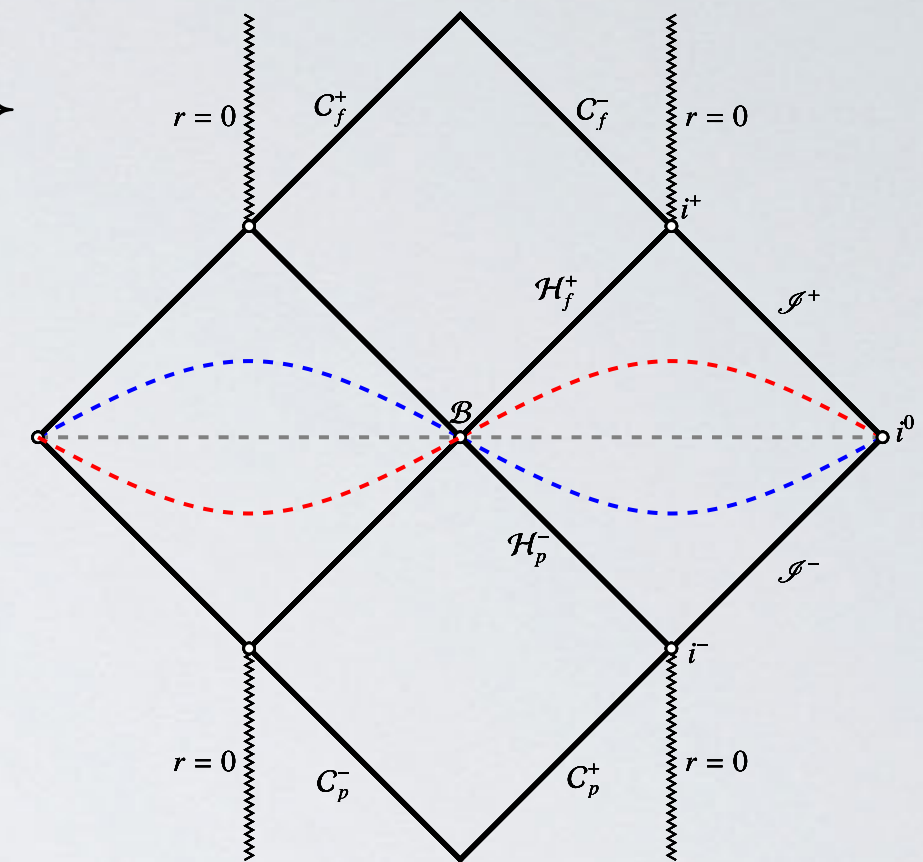
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$$\text{Im}(\omega_n) < 0$$



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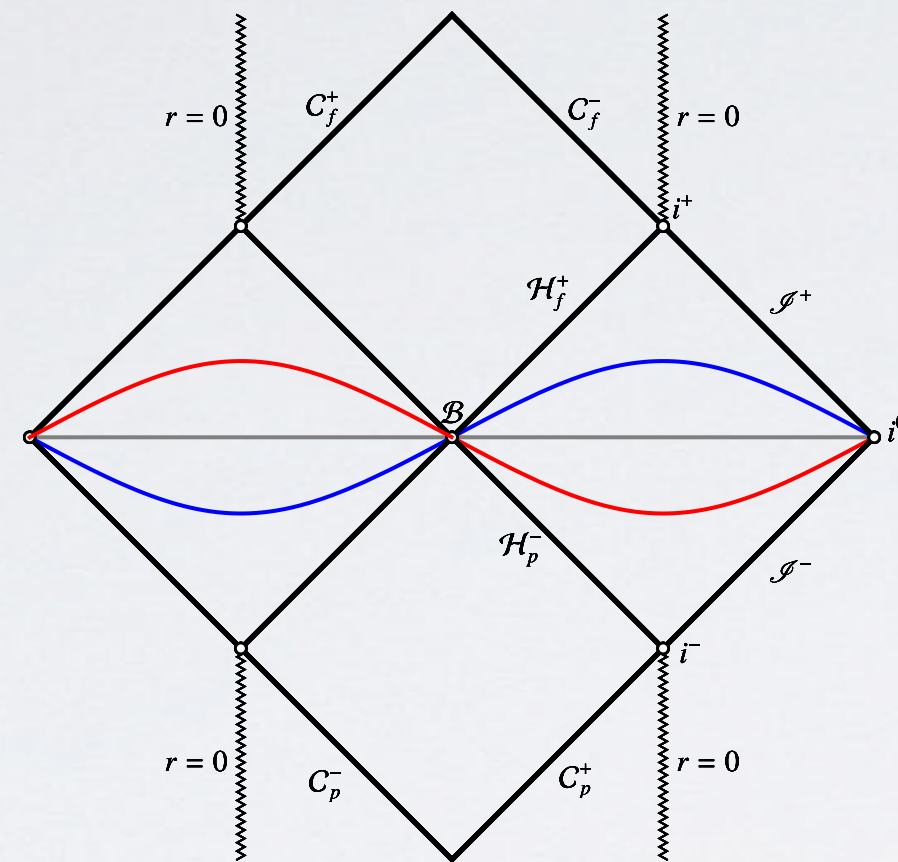


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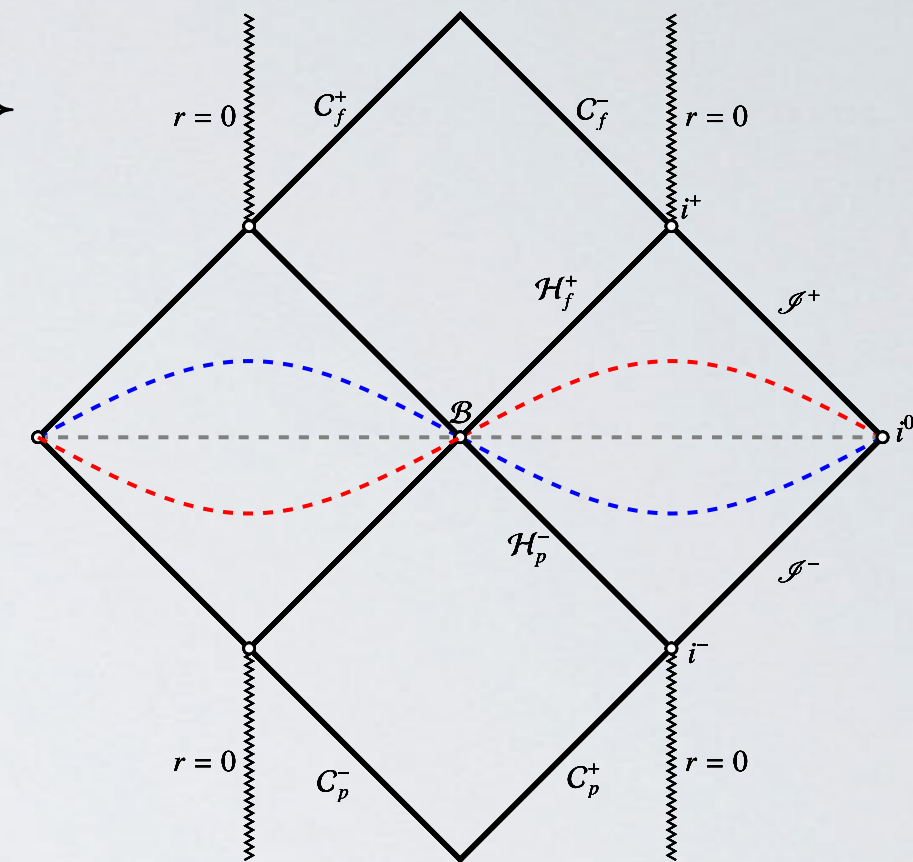
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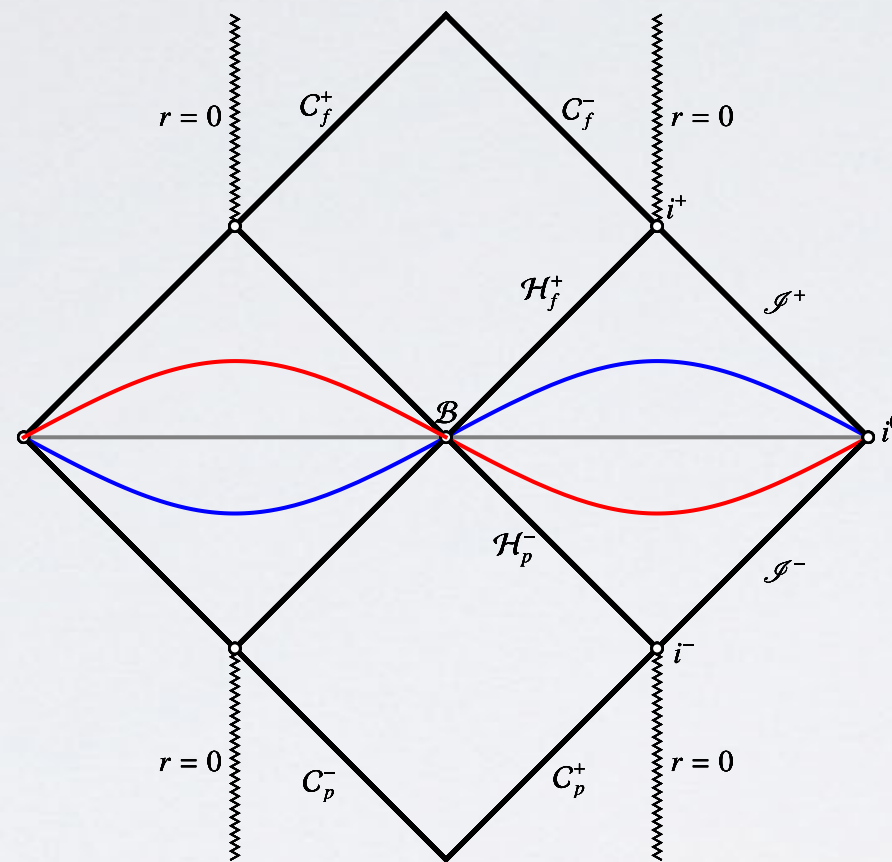
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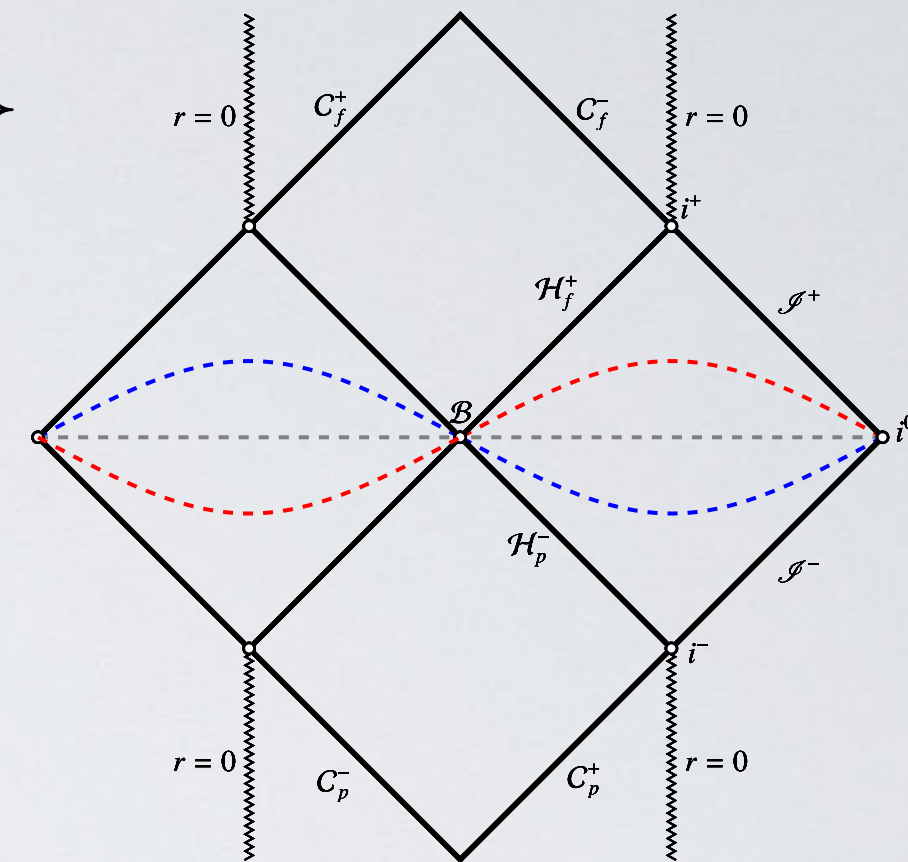
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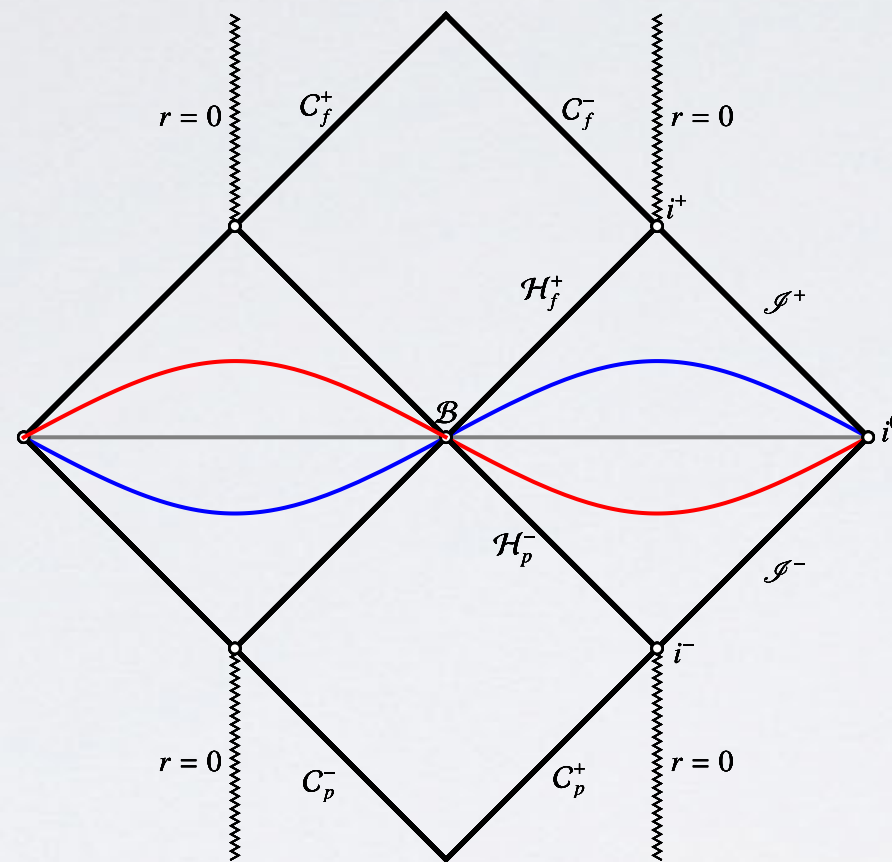
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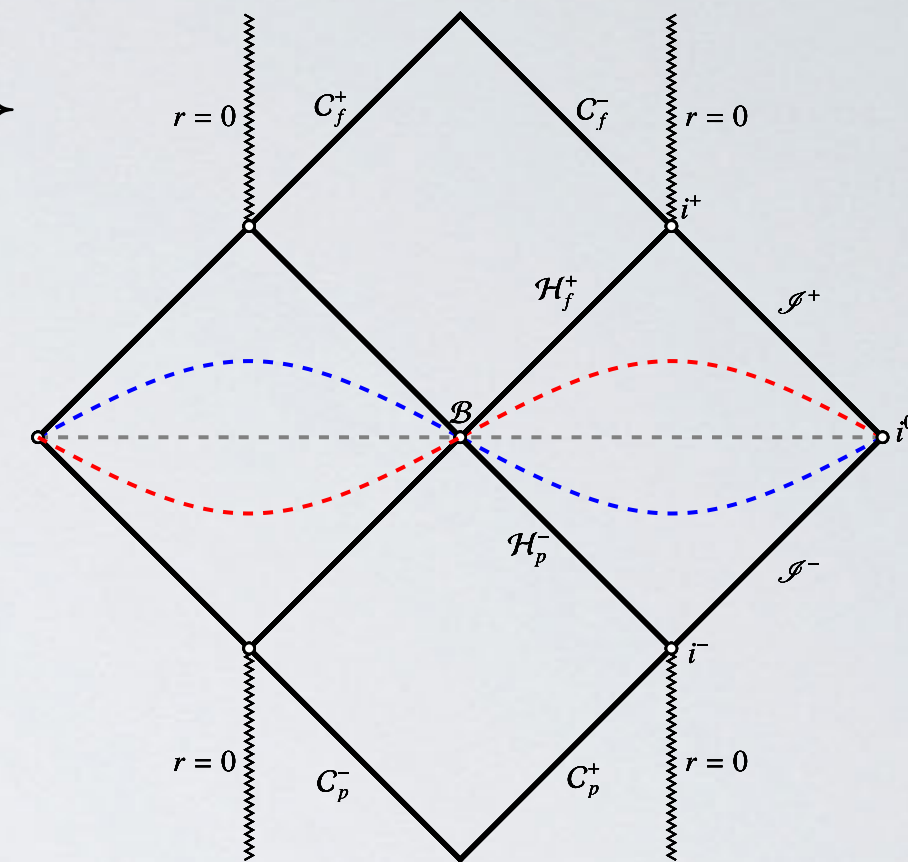
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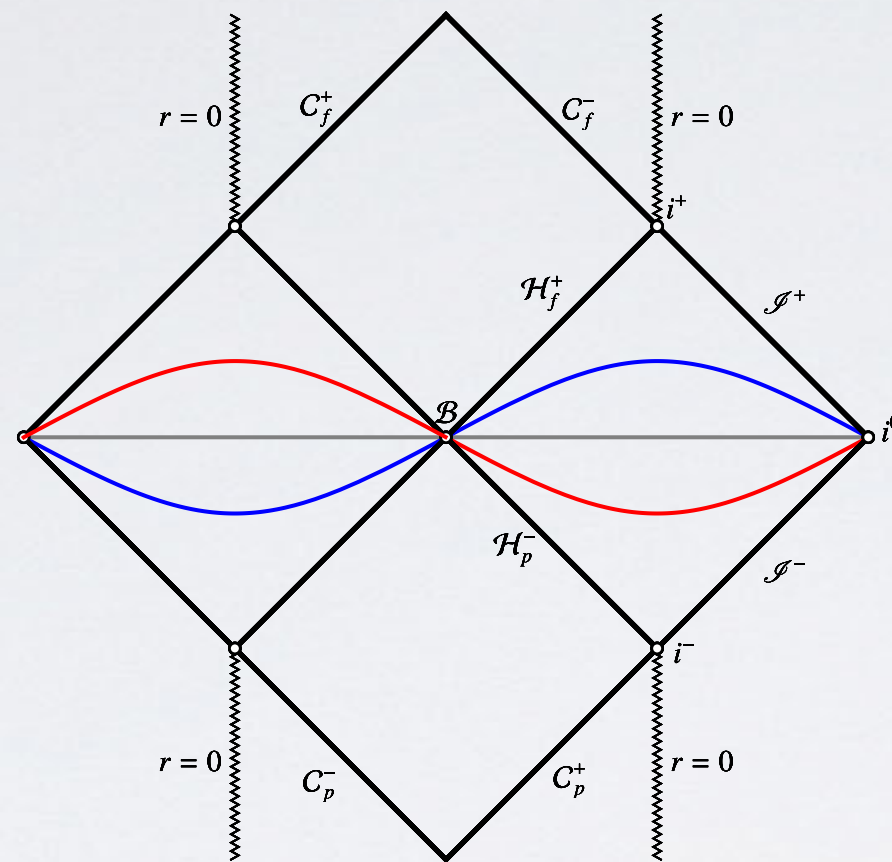
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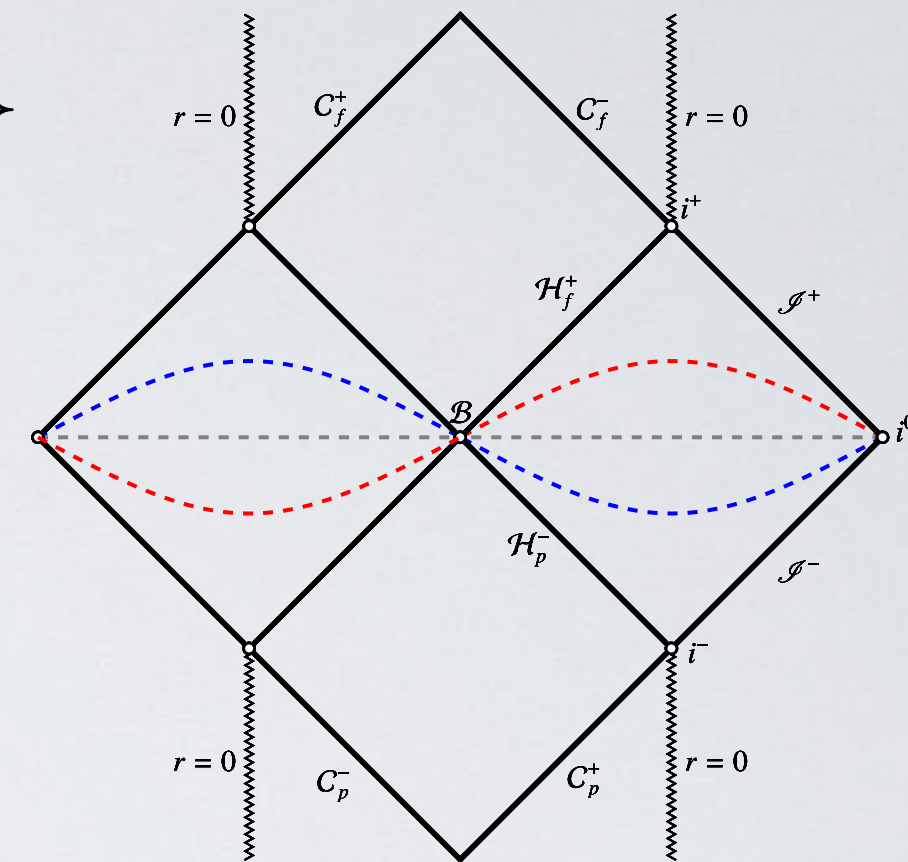
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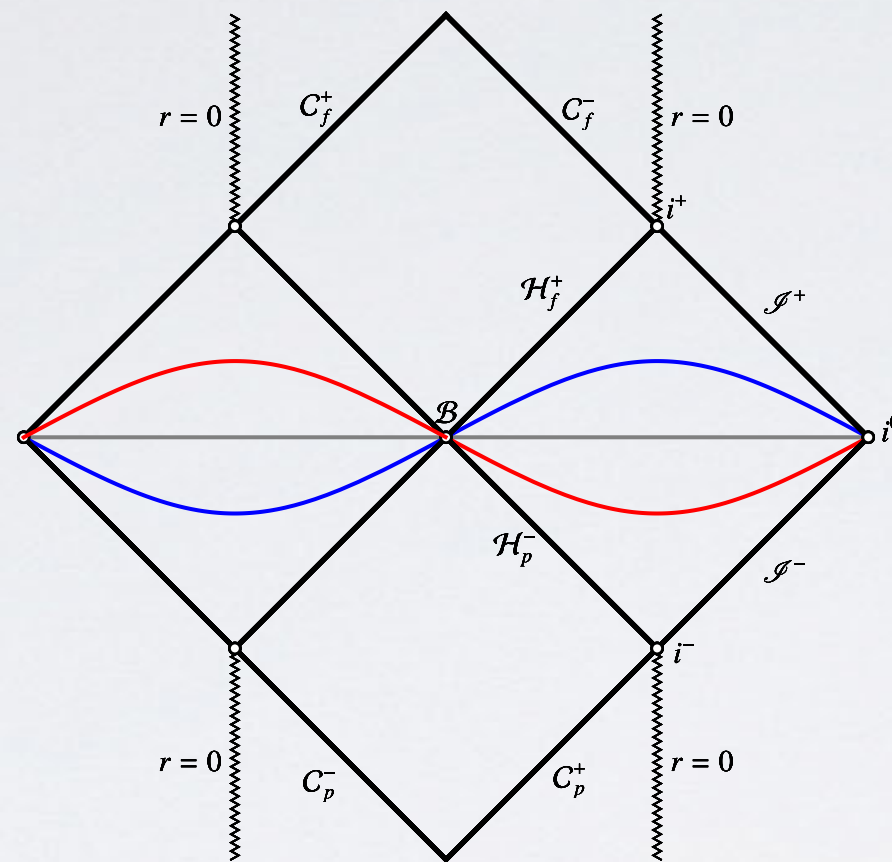
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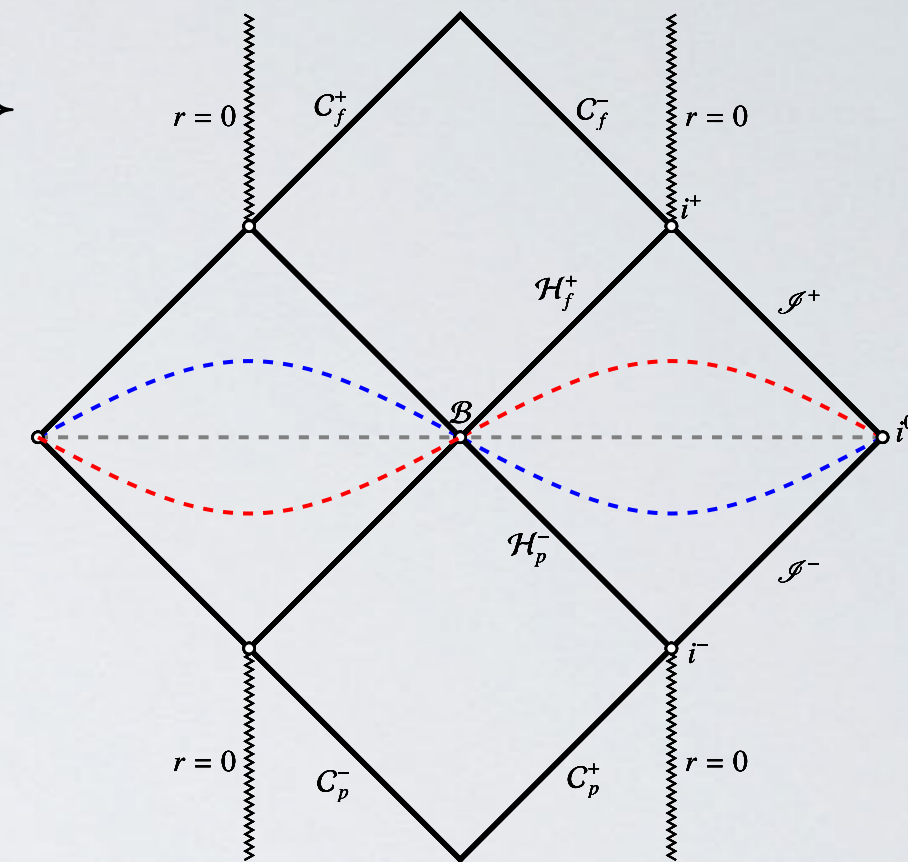


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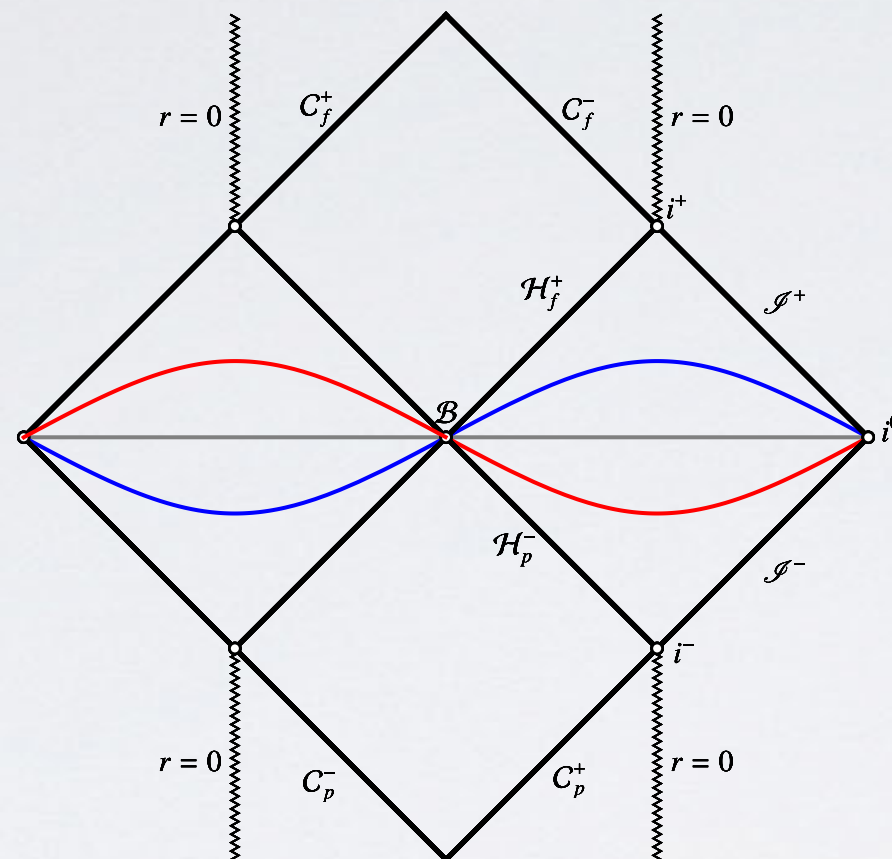
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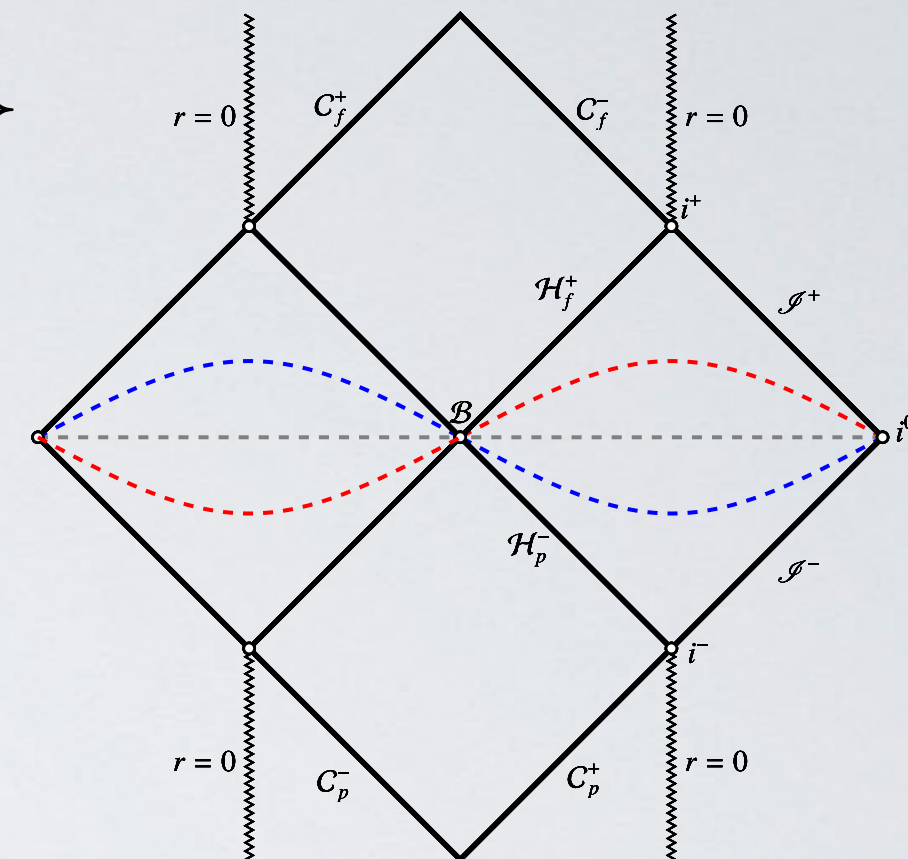


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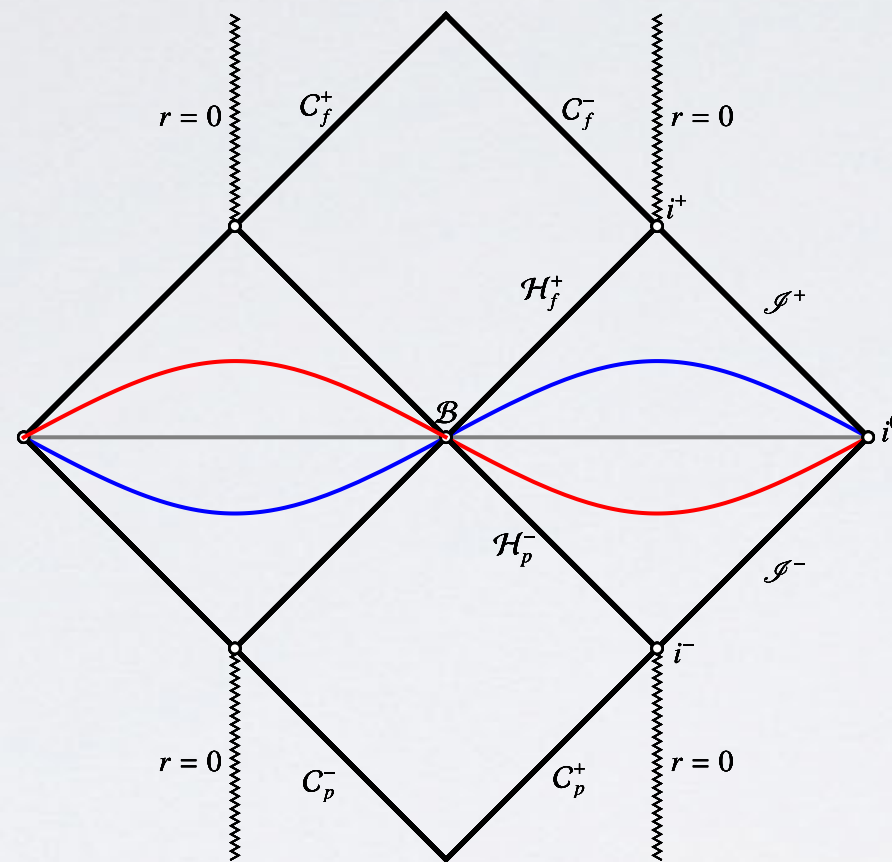
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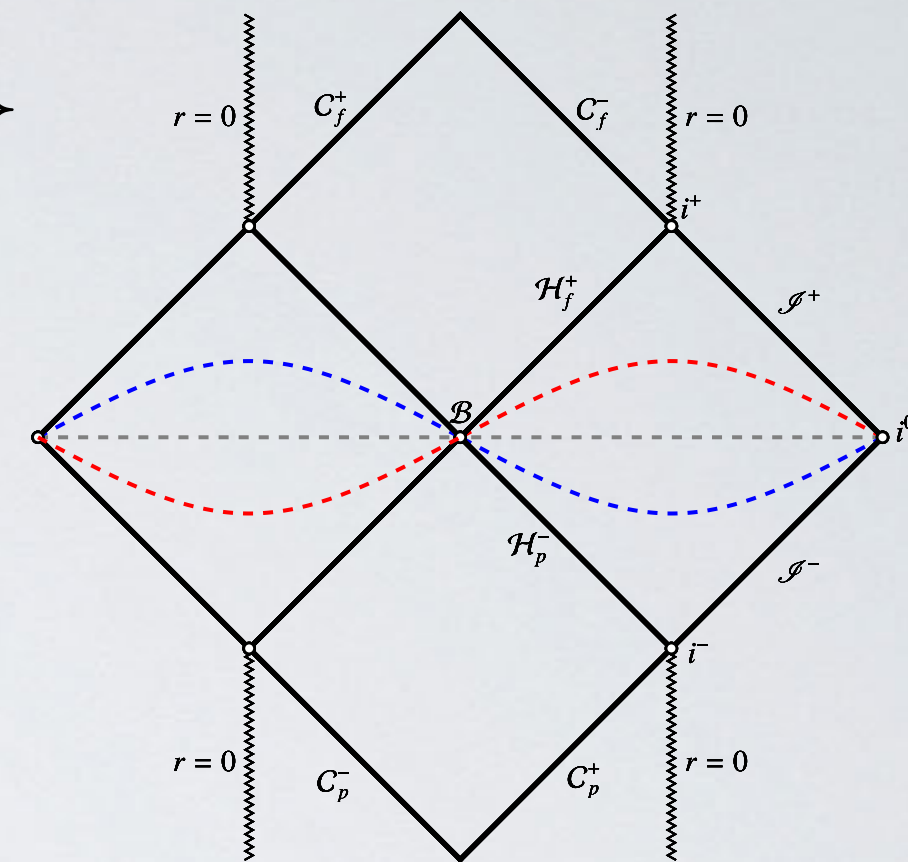


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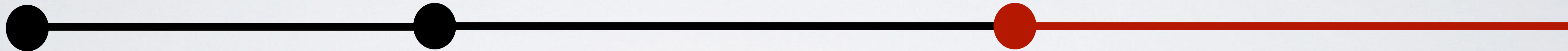
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OUTLINE

QNM products

$t-\phi$ 'Symmetry'

Future/past hyperboloidal



FUTURE / PAST - HYPERBOLOIDAL

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- From Schwarzschild $x^\mu = \{t, r\}$ hyperboloidal coordinates $\bar{x}_\pm^\mu = \{\tau_\pm, \sigma_\pm\}$

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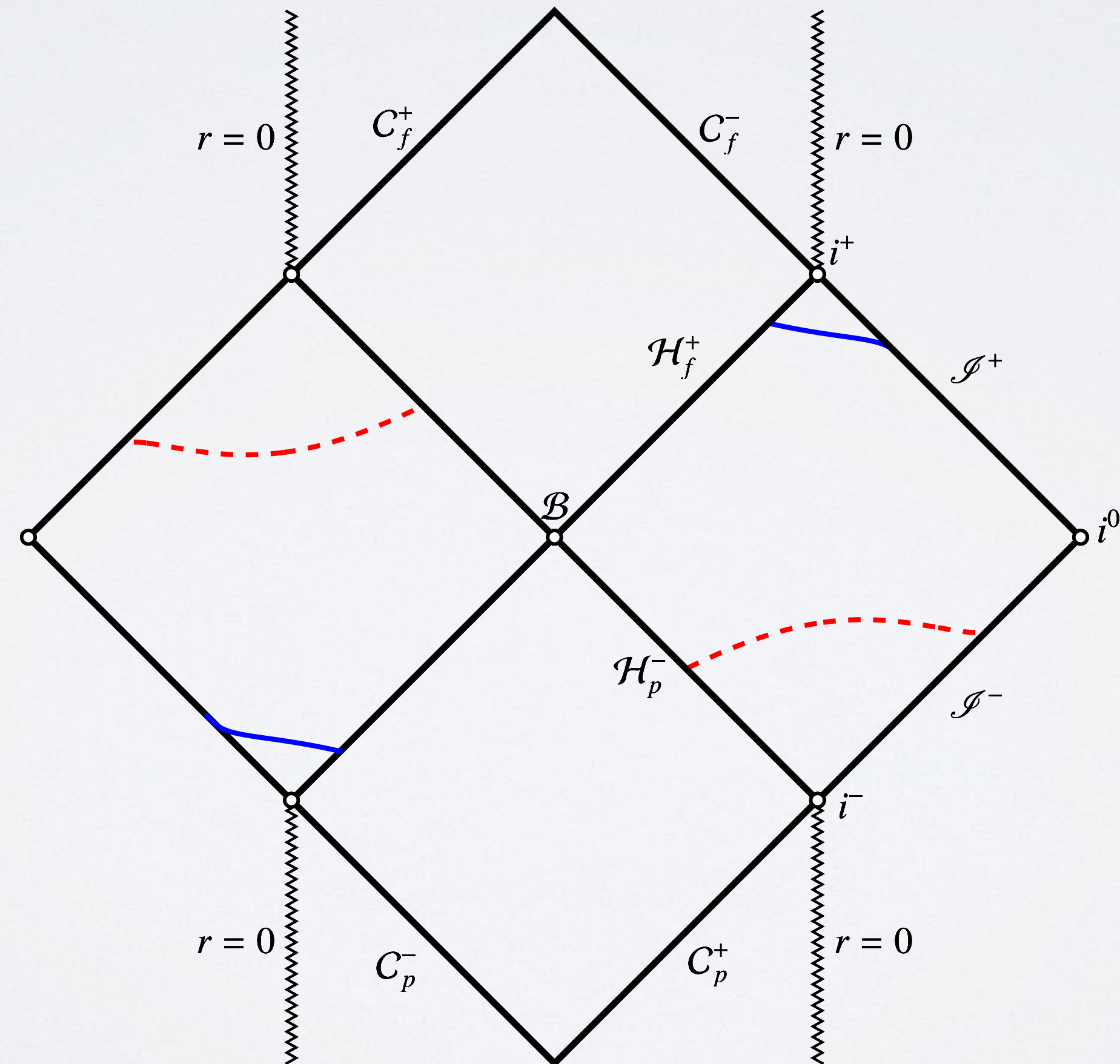
Metric is not invariant: $\tau_+ \leftrightarrow \tau_-$ is not a 'symmetry'

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$$\text{—} \quad \tau_+ = 2r_h$$

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- From hyperboloidal coordinates $\bar{x}_{\pm}^{\mu} = \{\tau_{\pm}, \sigma_{\pm}\}$ to their “time-flipped” $\check{x}_{\pm}^{\mu} = \{\check{\tau}_{\pm}, \check{\sigma}_{\pm}\}$

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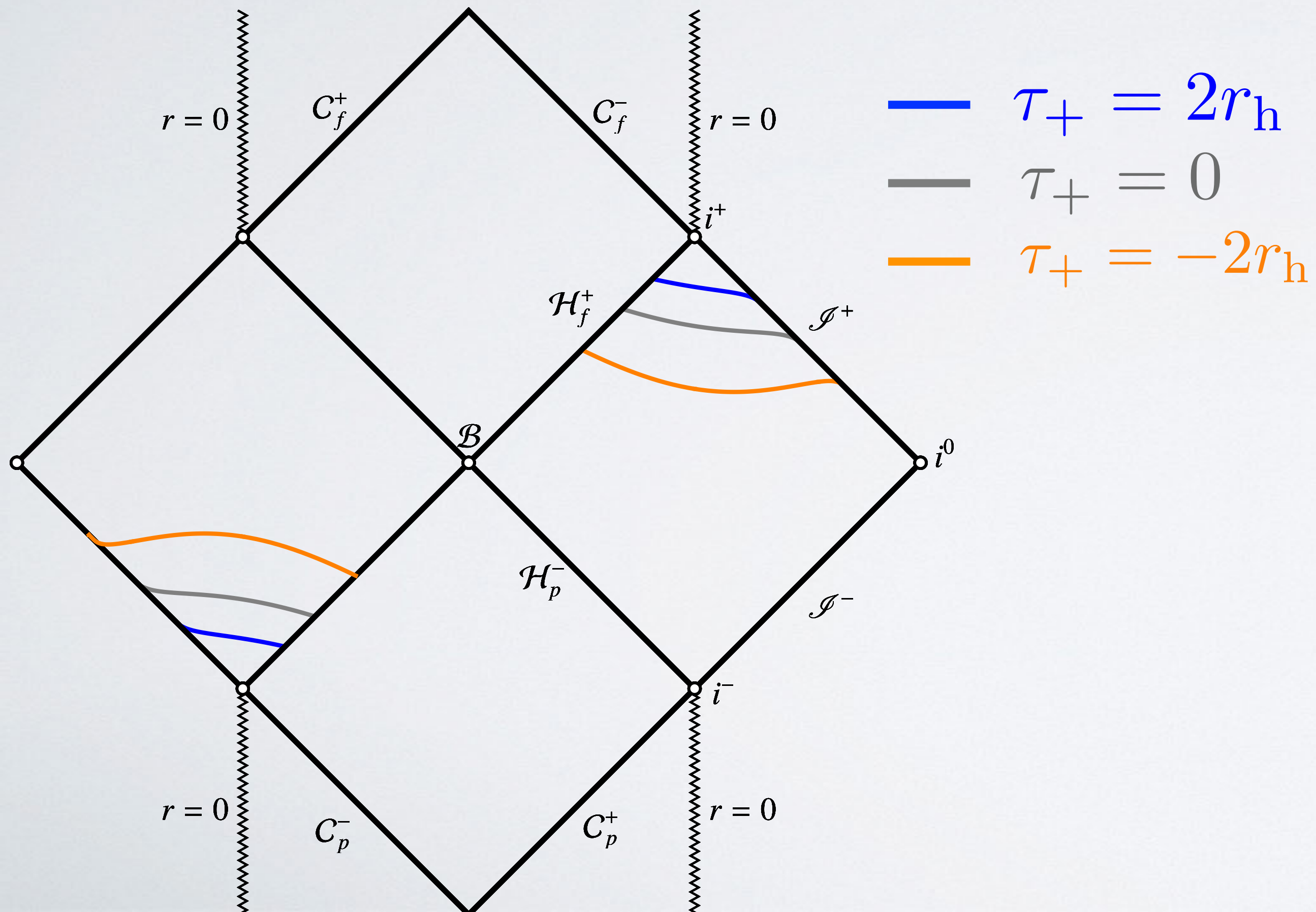
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TIME FLIP - HYPERBOLOIDAL

Future hyperboloids:

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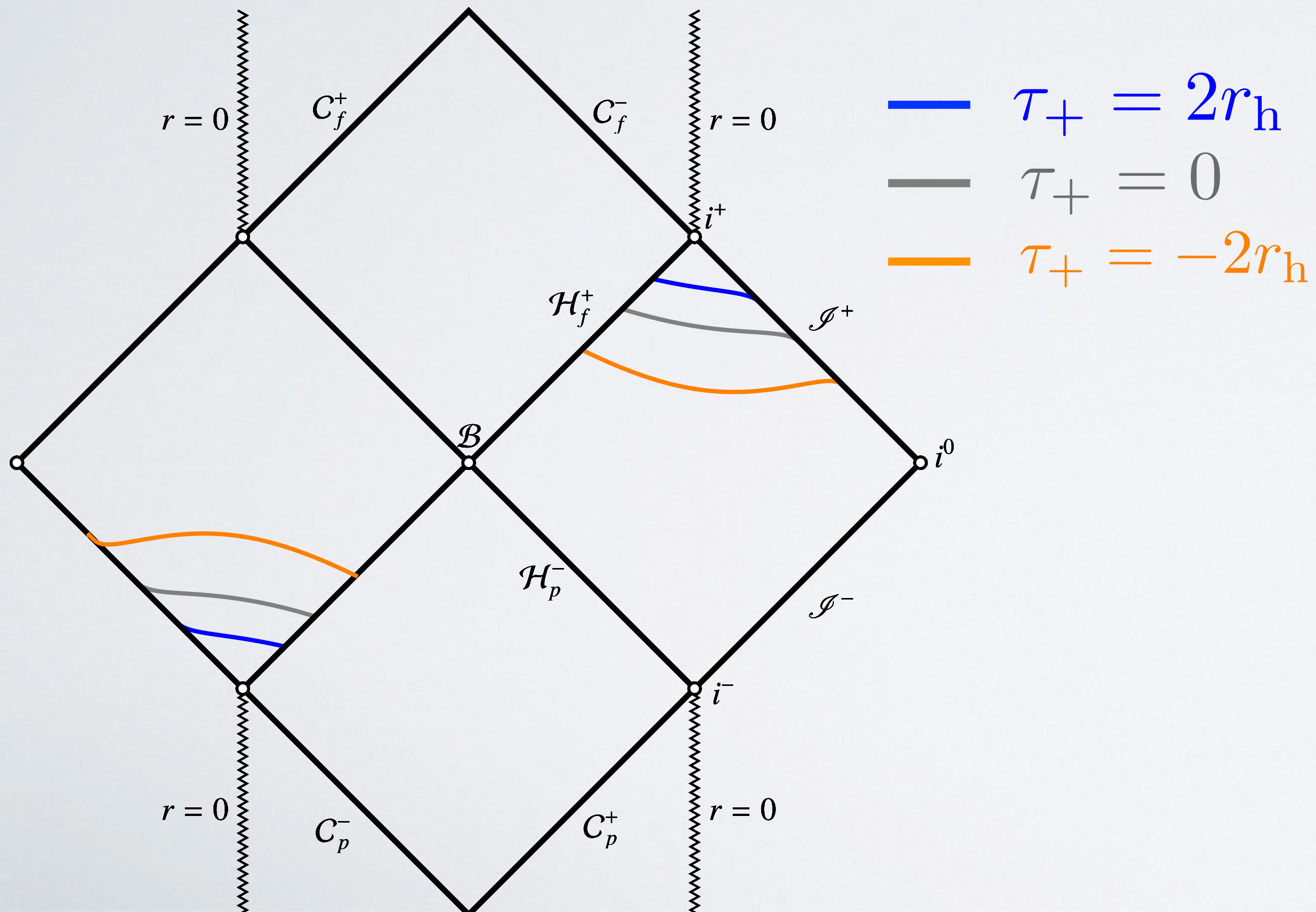


TIME FLIP - HYPERBOLOIDAL

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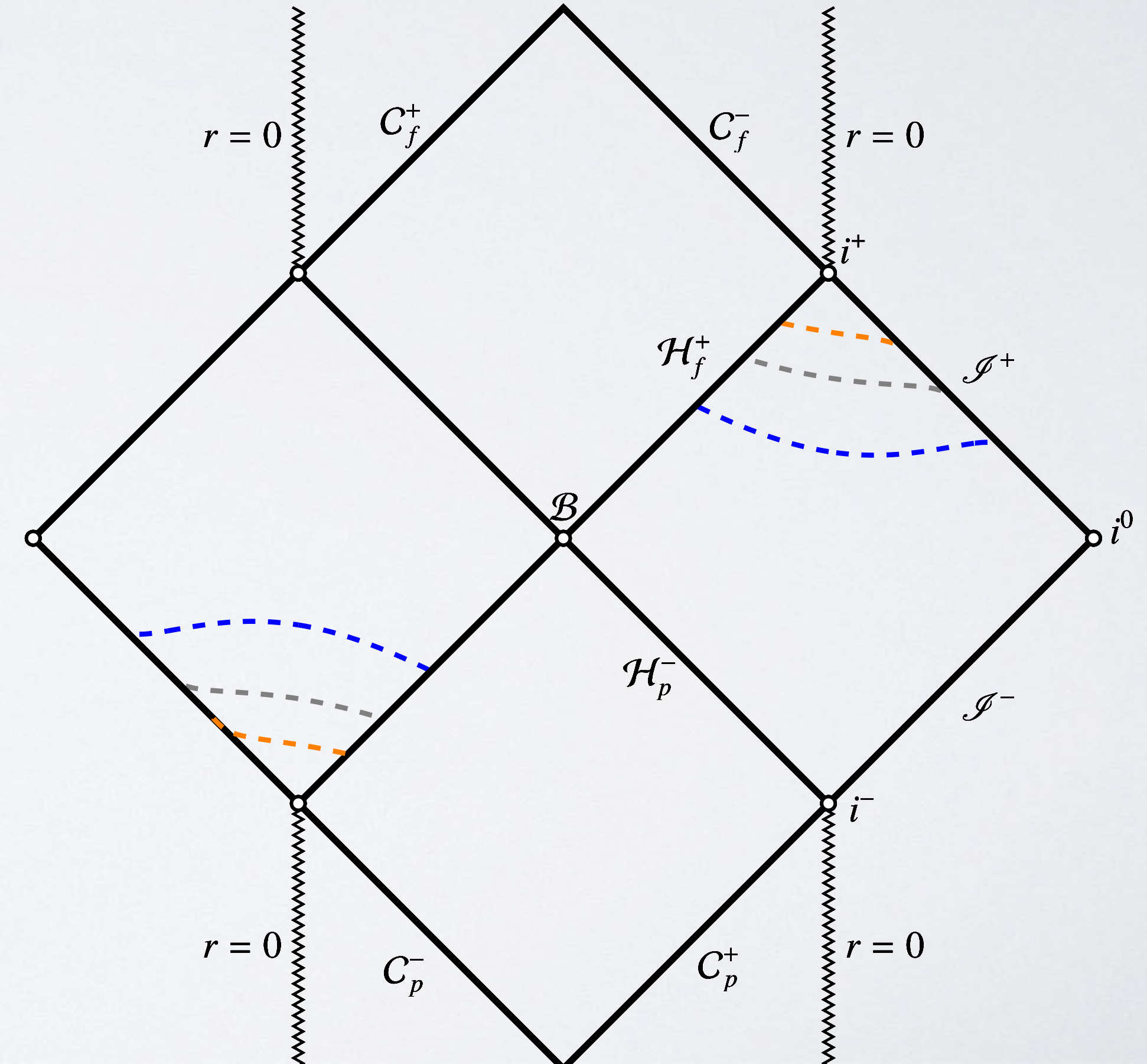
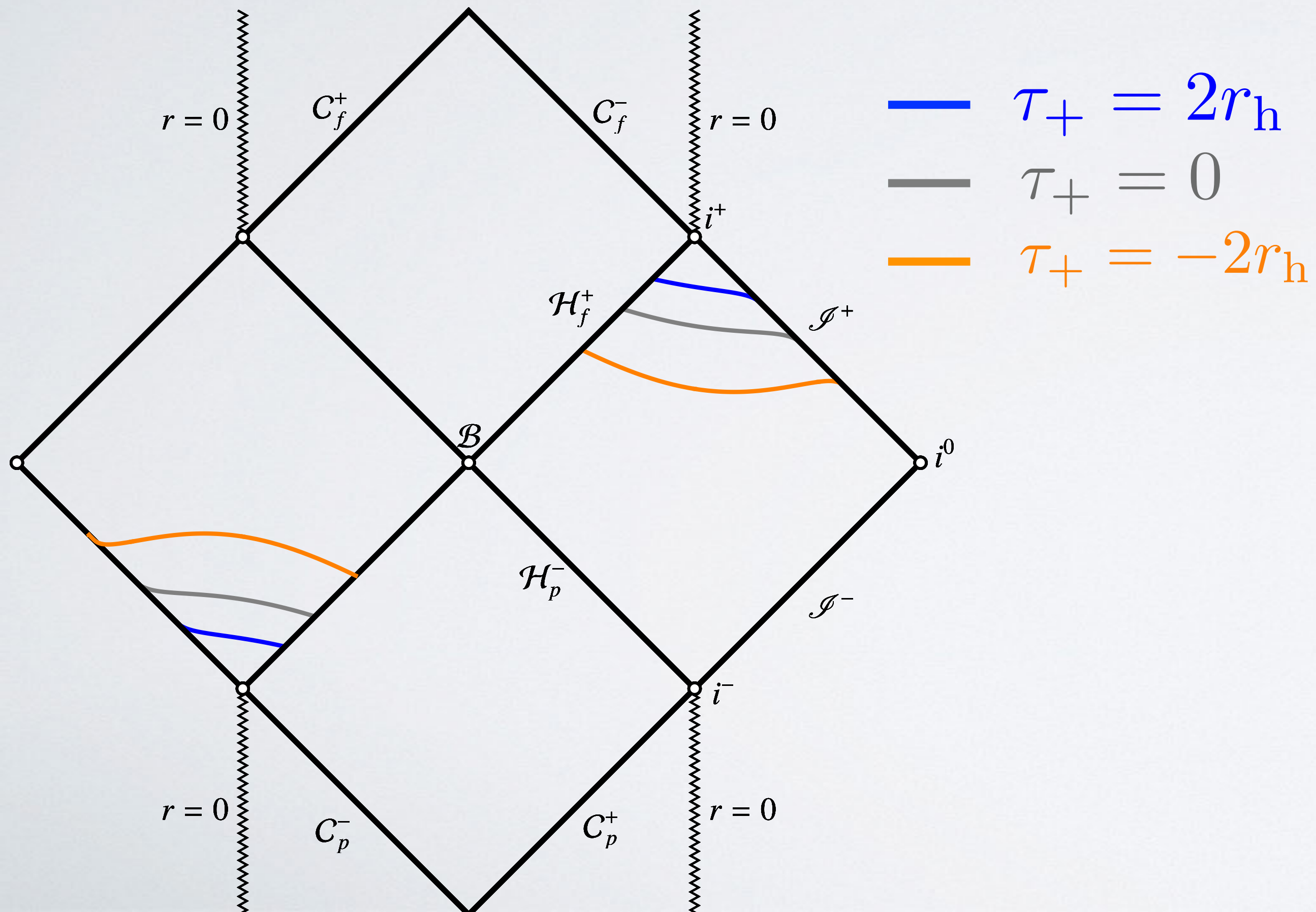


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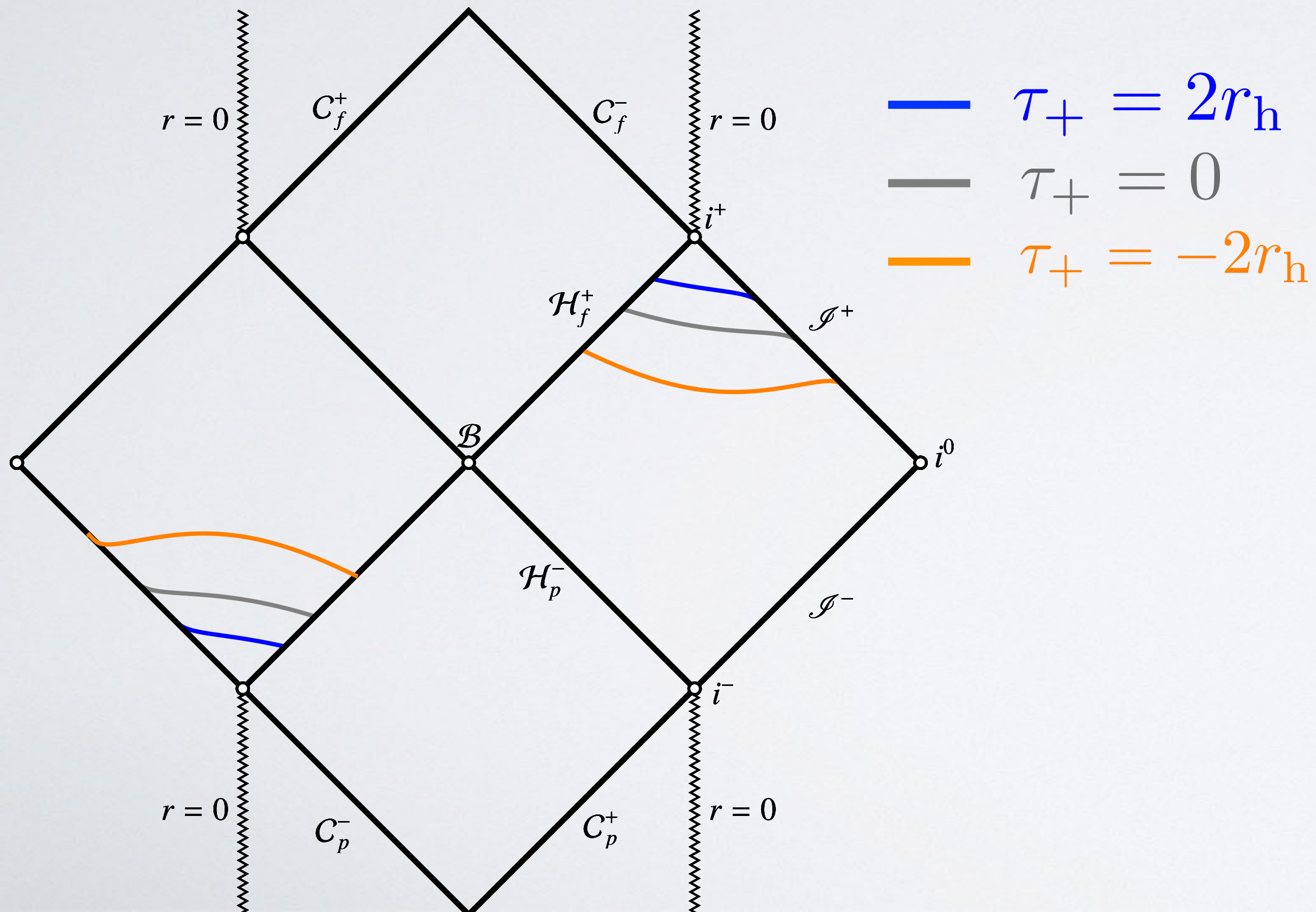
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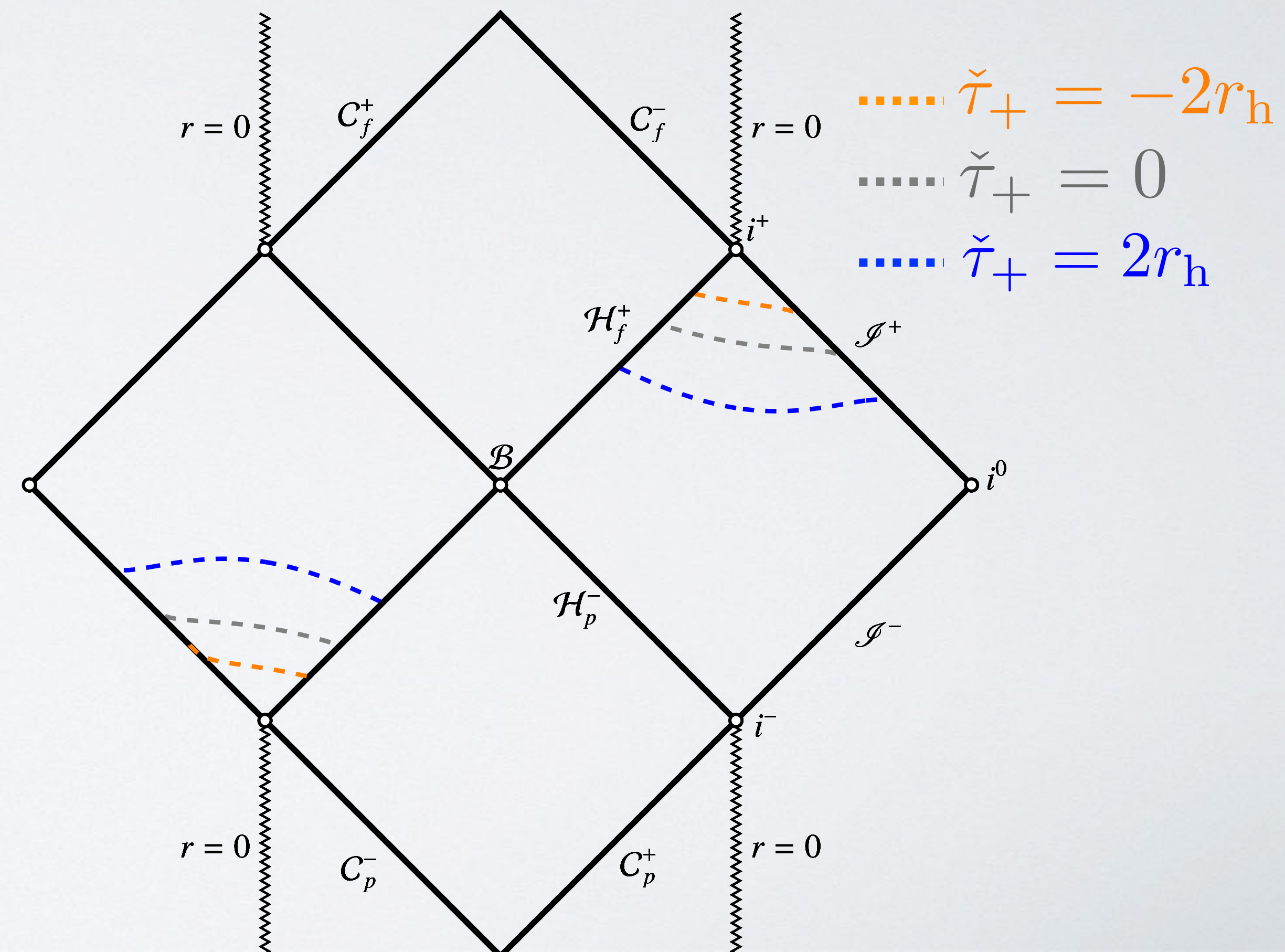
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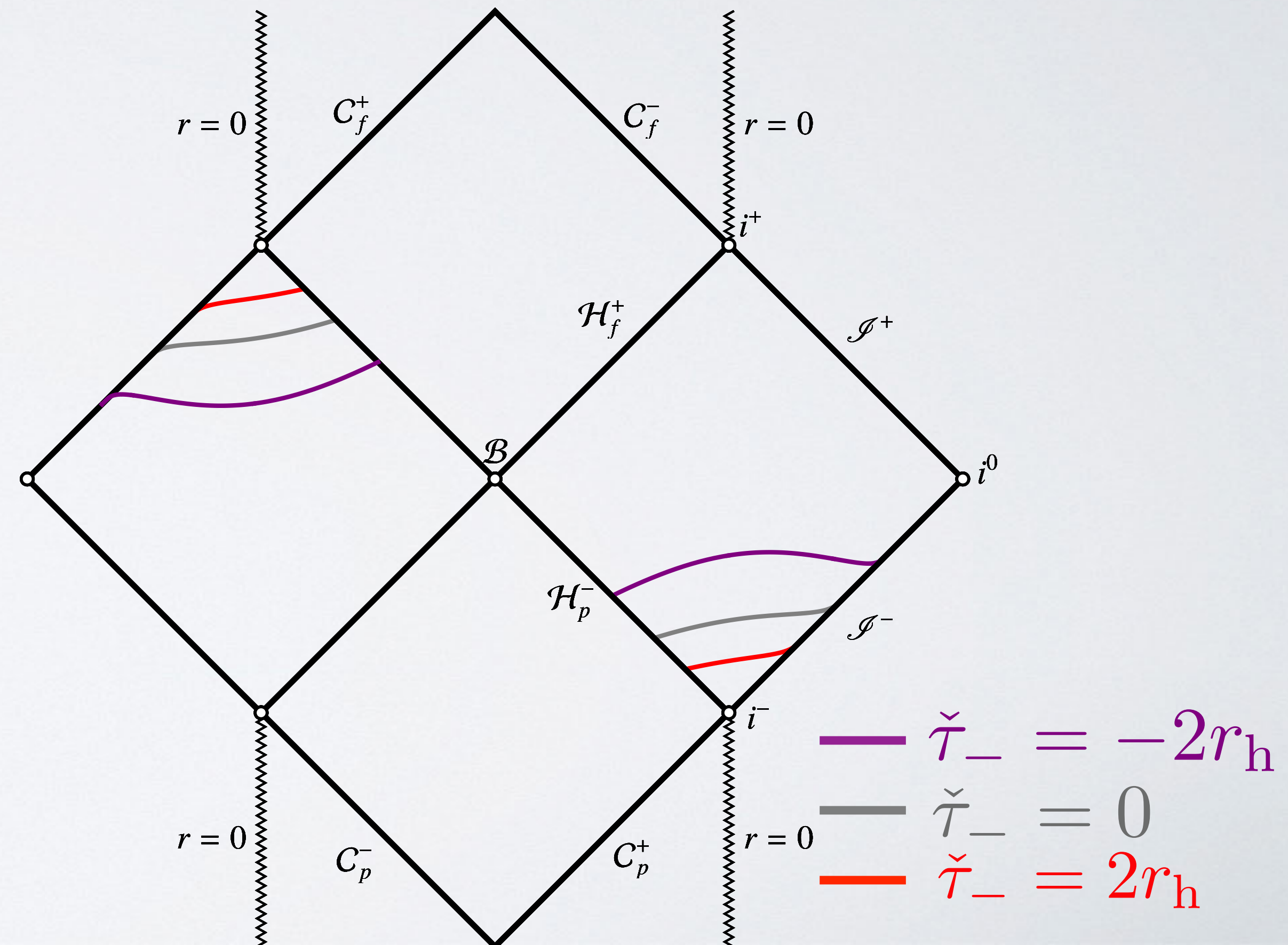
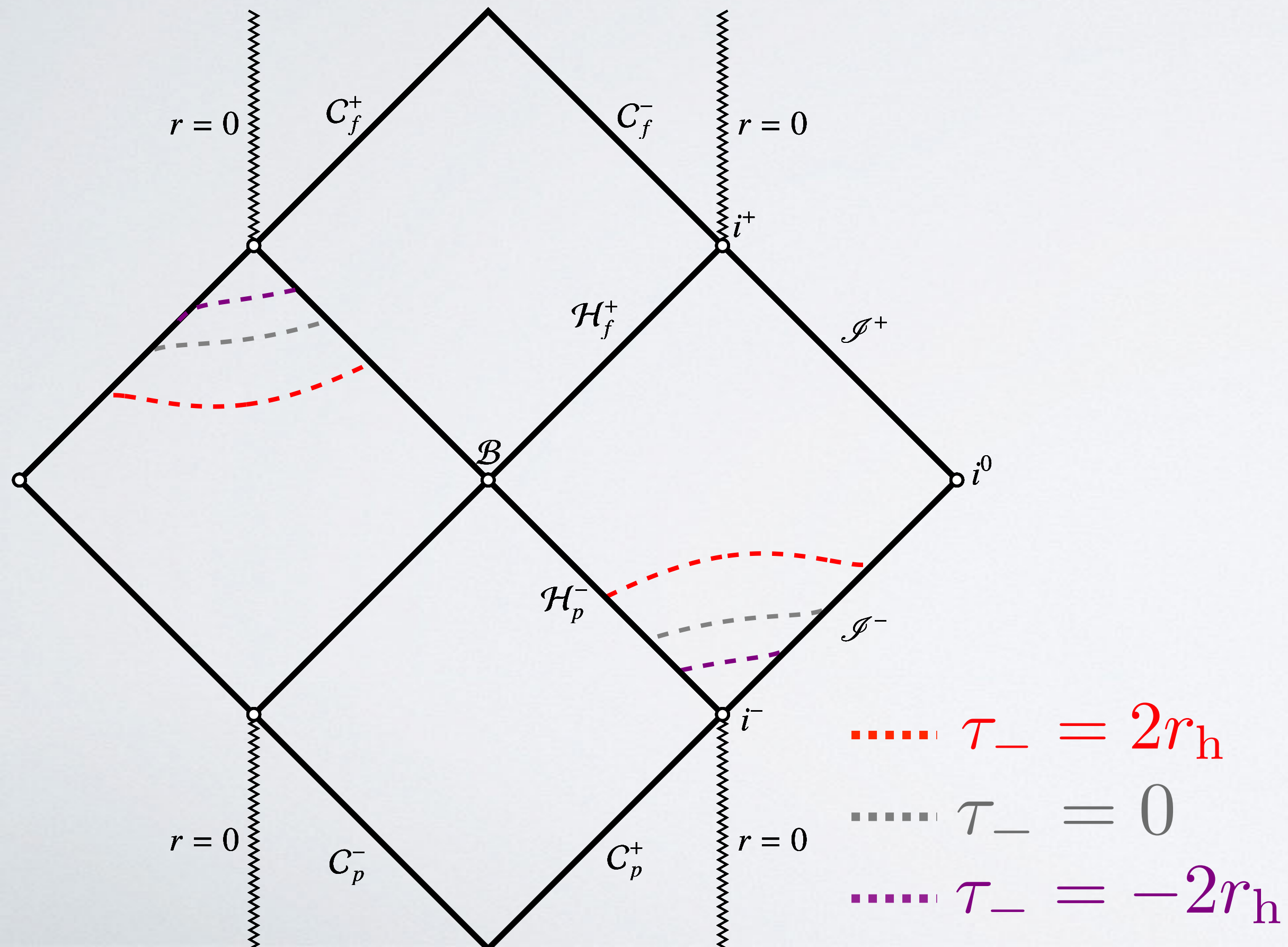


TIME FLIP - HYPERBOLOIDAL

Past hyperboloidals:

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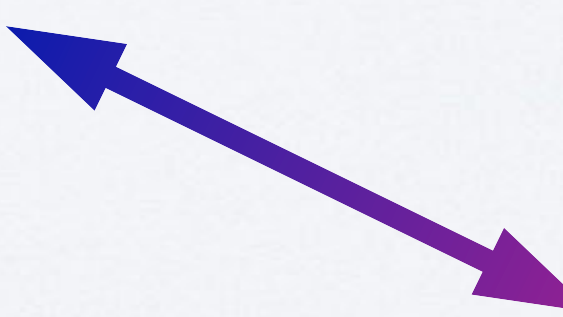
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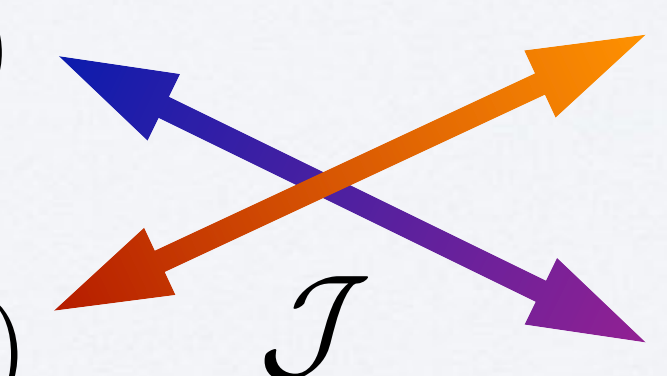
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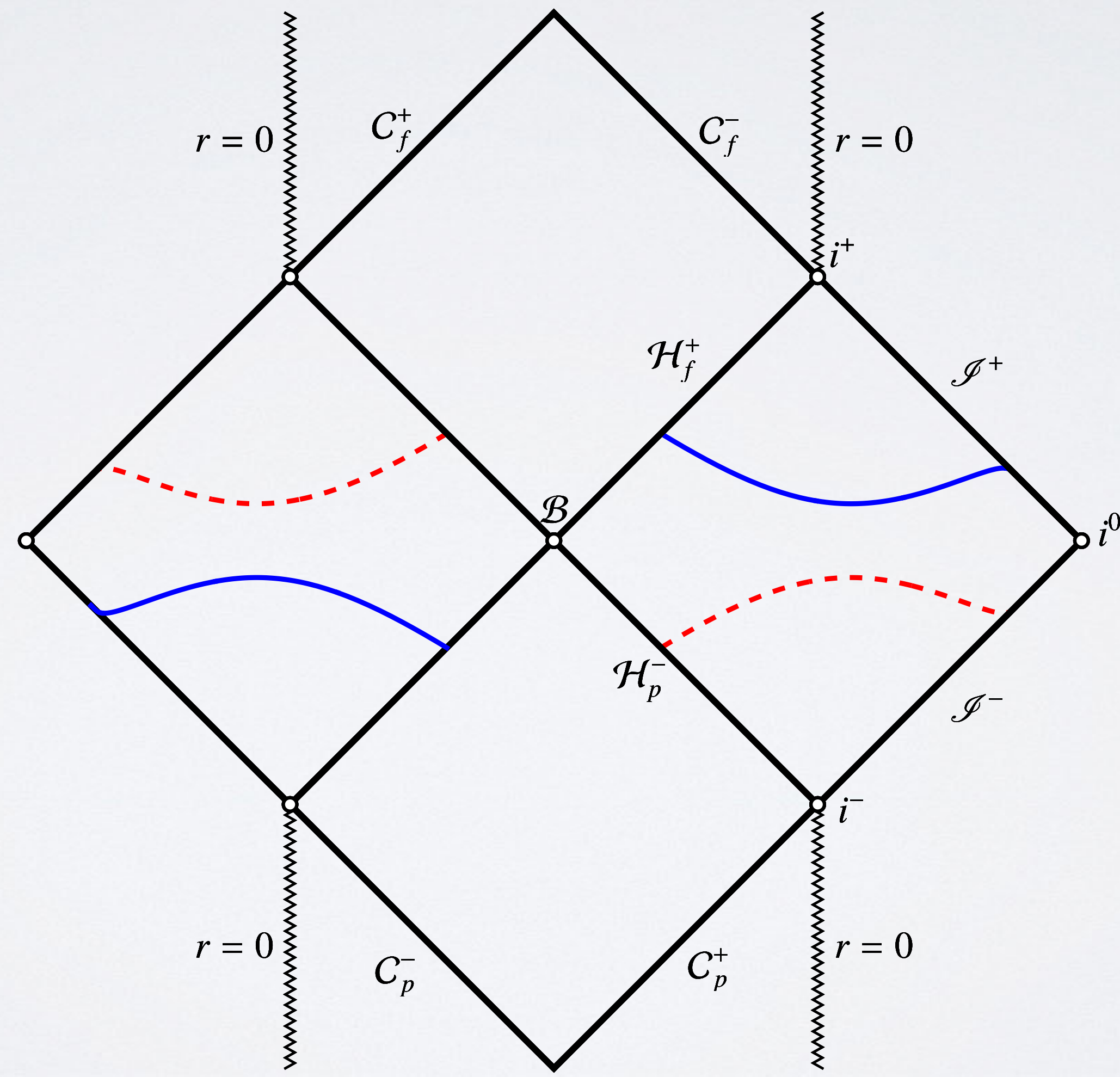
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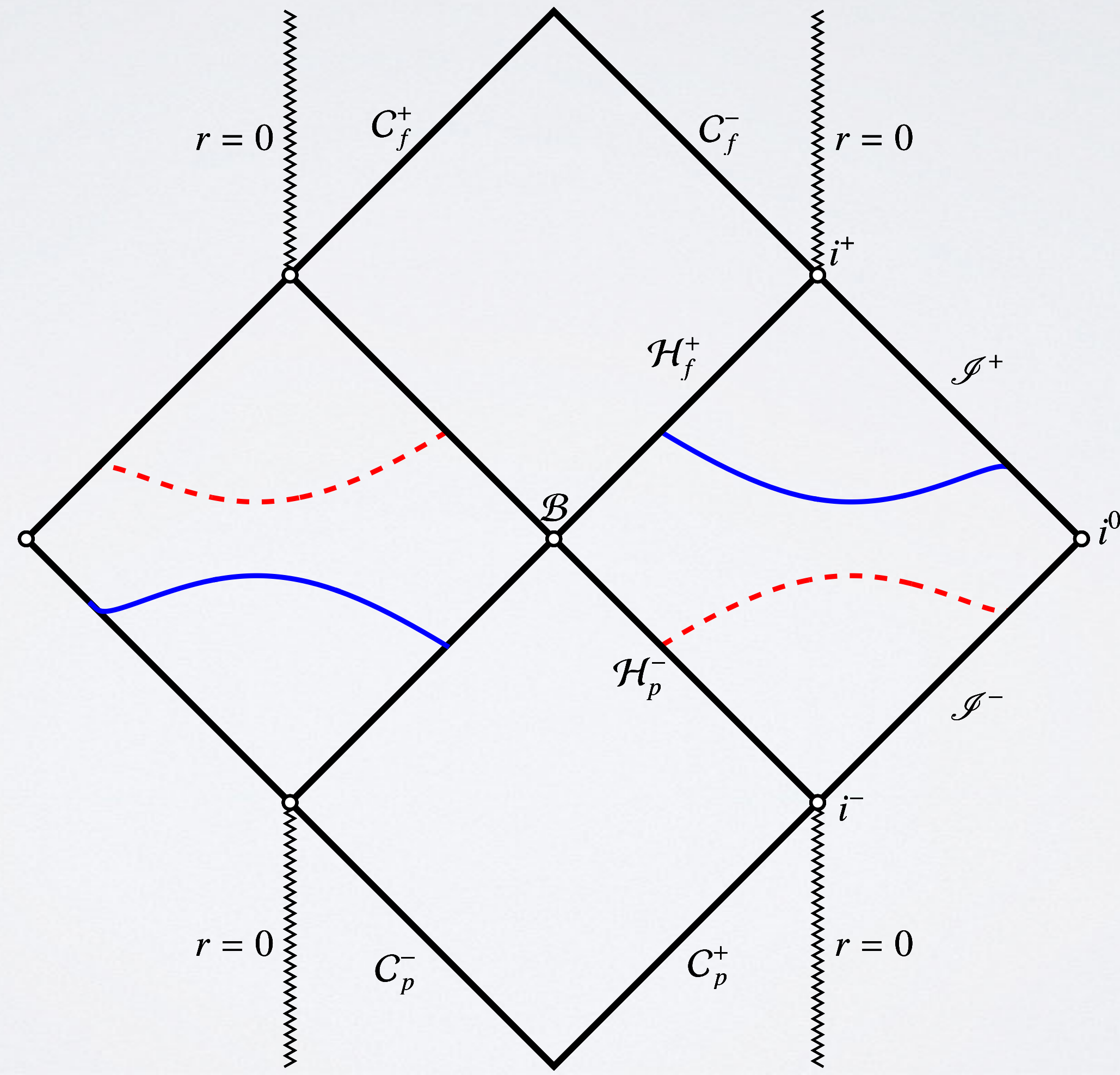
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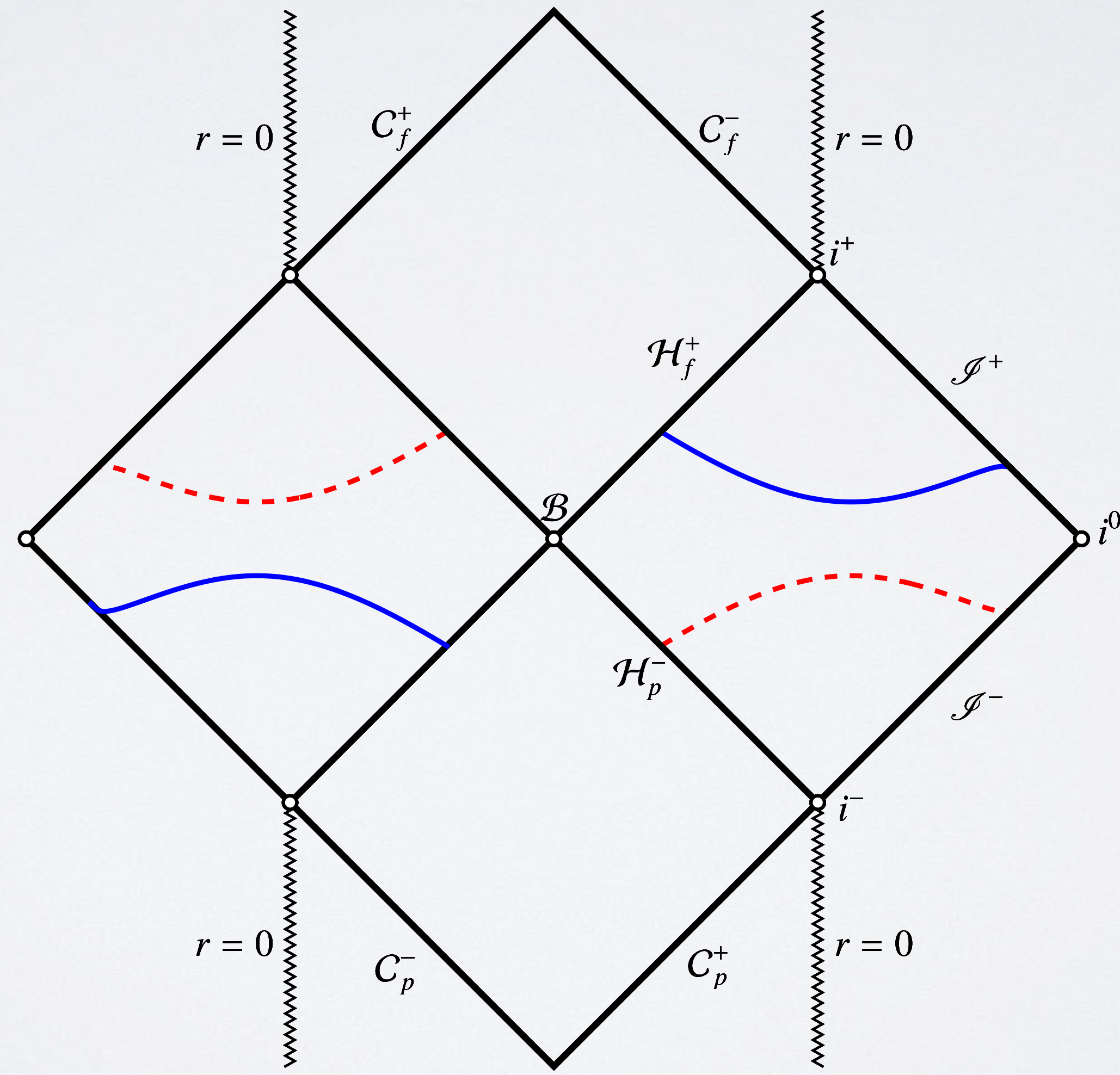
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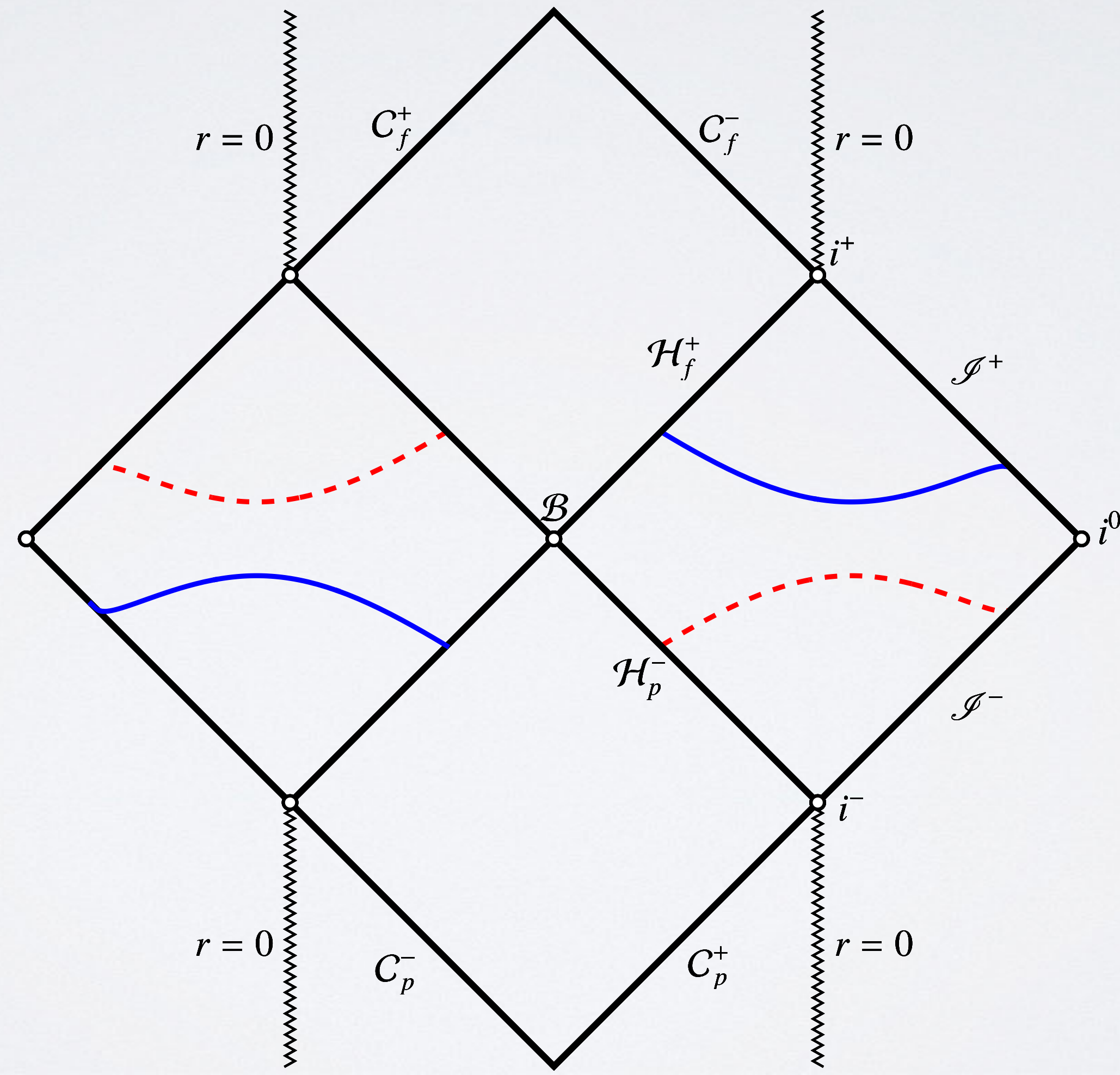
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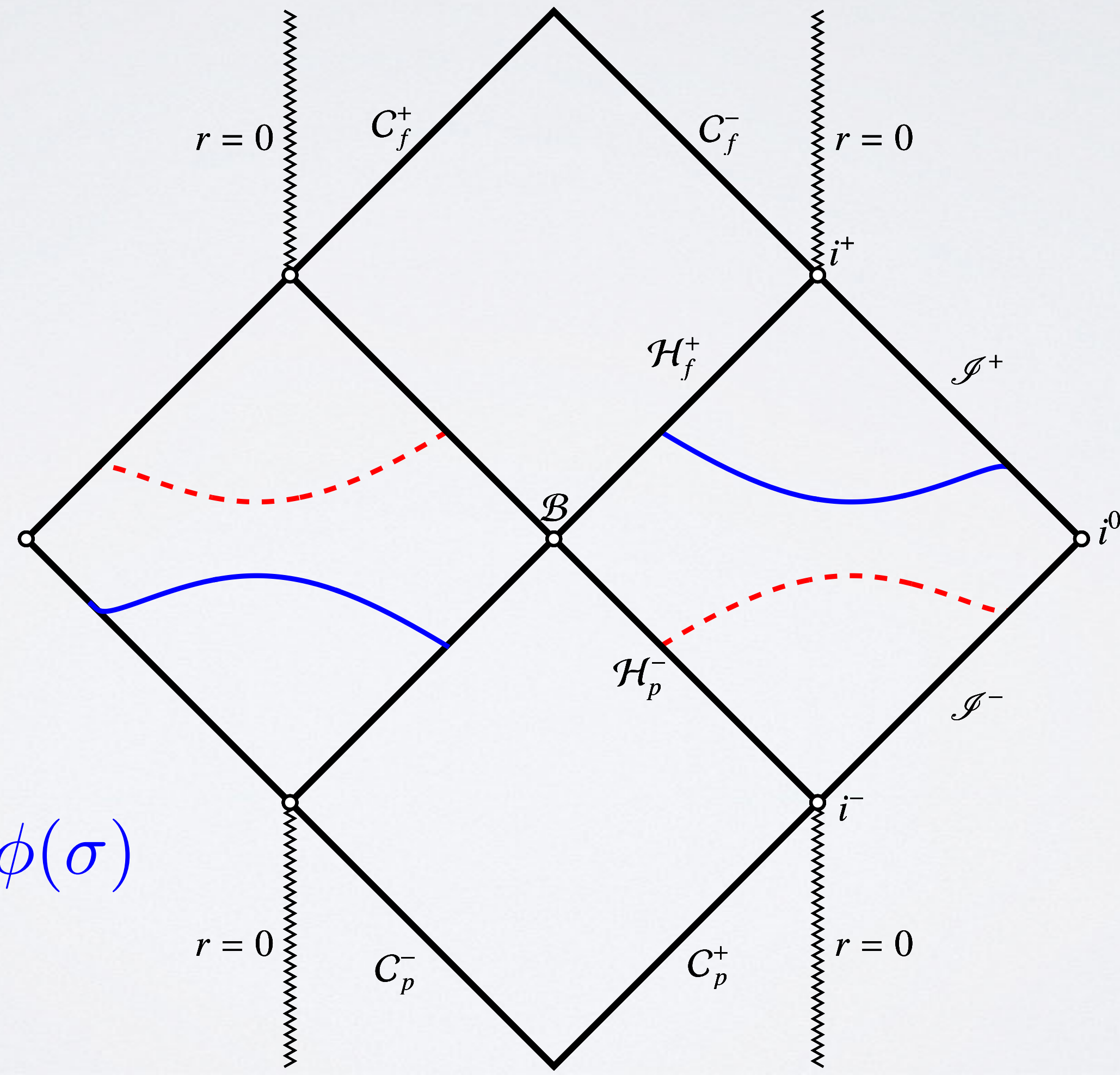
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$$\Phi^{\text{adv}}(\bar{x}^\mu) = e^{i\omega\tau} e^{-2i\omega H(\sigma)} \phi(\sigma)$$



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CONCLUSION

- Introduce time reversal operation in the hyperboloidal framework via future/past hyperboloidal foliations
- Motivation: particular strategy to construct QNMs bi-orthogonality products
- Product pairs QNM with anti-QNM eigenfunctions.
- Therefore, hyperboloidal framework does not remove blow-up factors